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Mao et al.

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(45) Date of Patent: Sep. 9, 2025

(54) **DIE ATTACHED LEVELING CONTROL BY METAL STOPPER BUMPS**

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(73) Assignee: **Taiwan Semiconductor Manufacturing Company, Ltd.**, Hsinchu (TW)

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Related U.S. Application Data

(63) Continuation of application No. 17/184,924, filed on Feb. 25, 2021, now Pat. No. 11,923,331.

(51) **Int. Cl.**
H01L 23/00 (2006.01)
H01L 21/48 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***H01L 24/32*** (2013.01); ***H01L 21/4803*** (2013.01); ***H01L 21/4853*** (2013.01);
(Continued)

(58) **Field of Classification Search**
None
See application file for complete search history.

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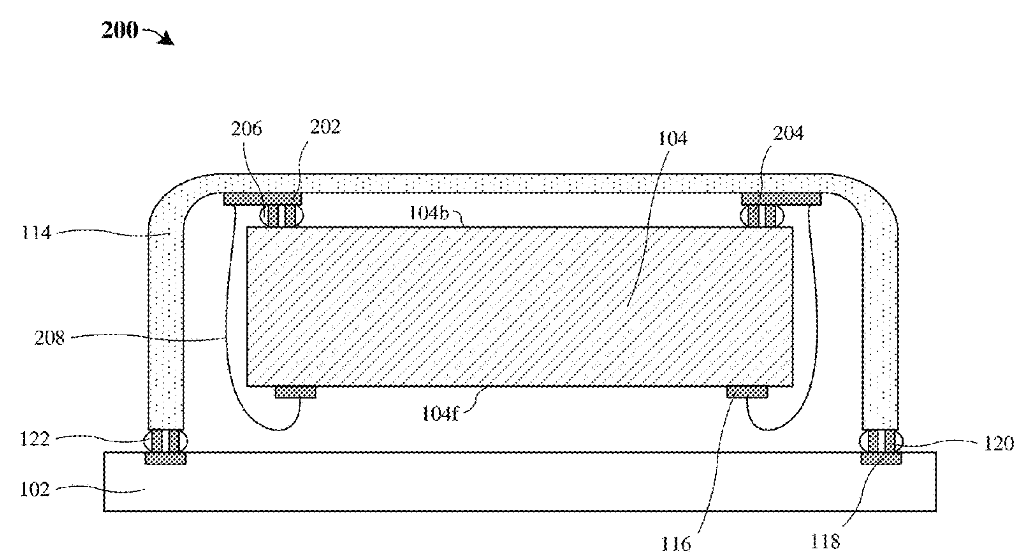
Primary Examiner — Telly D Green
(74) *Attorney, Agent, or Firm* — Eschweiler & Potashnik, LLC

(57) **ABSTRACT**

In some embodiments, the present disclosure relates to an integrated chip including a substrate and a first die disposed over the substrate. A first plurality of die stopper bumps are disposed along a backside of the first die. The first plurality of die stopper bumps directly contact the backside of the first die, and the first plurality of die stopper bumps are arranged as a plurality of groups of die stopper bumps. A plurality of adhesive structures are also present. Each of the plurality of adhesive structures surrounds a corresponding group of the plurality of groups of die stopper bumps.

20 Claims, 32 Drawing Sheets



(51) **Int. Cl.**

H01L 23/10	(2006.01)
H01L 23/498	(2006.01)
H01L 25/00	(2006.01)
H01L 25/065	(2023.01)

(52) **U.S. Cl.**

CPC ***H01L 23/10*** (2013.01); ***H01L 23/49838*** (2013.01); ***H01L 24/14*** (2013.01); ***H01L 24/16*** (2013.01); ***H01L 24/17*** (2013.01); ***H01L 24/33*** (2013.01); ***H01L 24/48*** (2013.01); ***H01L 24/73*** (2013.01); ***H01L 24/81*** (2013.01); ***H01L 24/83*** (2013.01); ***H01L 24/92*** (2013.01); ***H01L 25/0657*** (2013.01); ***H01L 25/50*** (2013.01); *H01L 24/05* (2013.01); *H01L 2224/05554* (2013.01); *H01L 2224/1403* (2013.01); *H01L 2224/1601* (2013.01); *H01L 2224/16014* (2013.01); *H01L 2224/16145* (2013.01); *H01L 2224/16227* (2013.01); *H01L 2224/1703* (2013.01); *H01L 2224/17106* (2013.01); *H01L 2224/17107* (2013.01); *H01L 2224/26155* (2013.01); *H01L 2224/32145* (2013.01); *H01L 2224/32227* (2013.01); *H01L 2224/3303* (2013.01); *H01L 2224/33181* (2013.01); *H01L 2224/48105* (2013.01); *H01L 2224/48145* (2013.01); *H01L 2224/48227* (2013.01); *H01L 2224/73104* (2013.01); *H01L 2224/73207* (2013.01); *H01L 2224/73257* (2013.01); *H01L 2224/81191* (2013.01); *H01L 2224/81192* (2013.01); *H01L 2224/819* (2013.01); *H01L 2224/83191* (2013.01); *H01L 2224/83192* (2013.01); *H01L 2224/9211* (2013.01); *H01L 2224/92247* (2013.01); *H01L 2225/0651* (2013.01); *H01L 2225/06513* (2013.01); *H01L 2225/06517* (2013.01); *H01L 2225/06568* (2013.01); *H01L 2225/06582* (2013.01)

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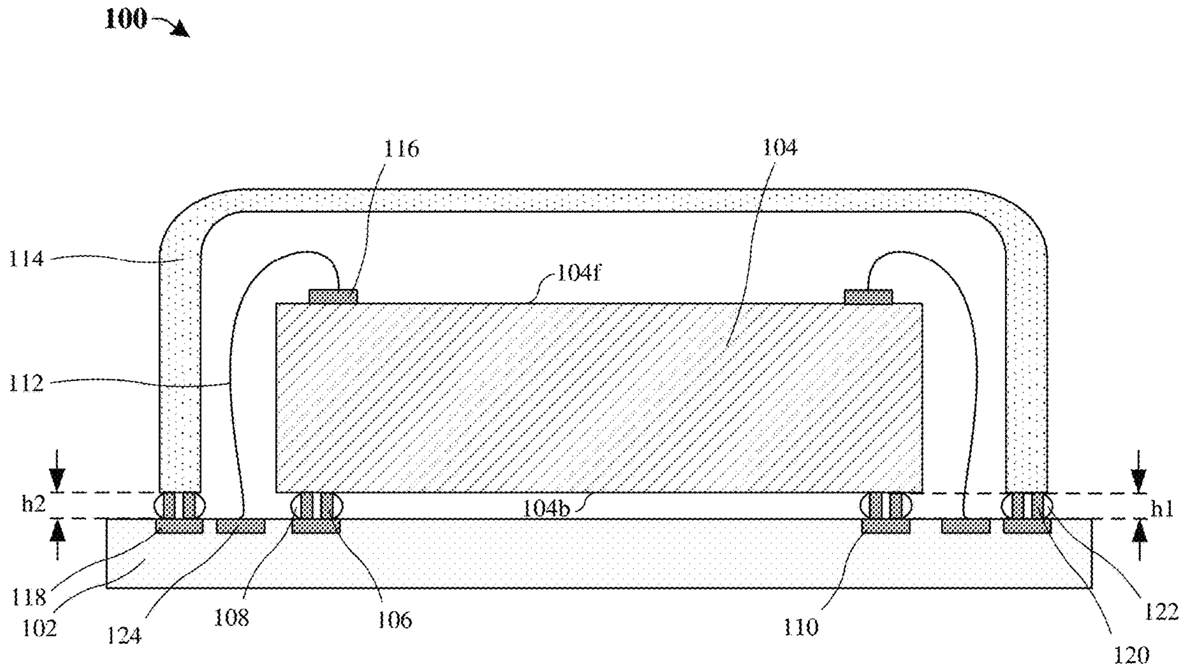
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116
104
114
104f
112
h2
104b
h1
118
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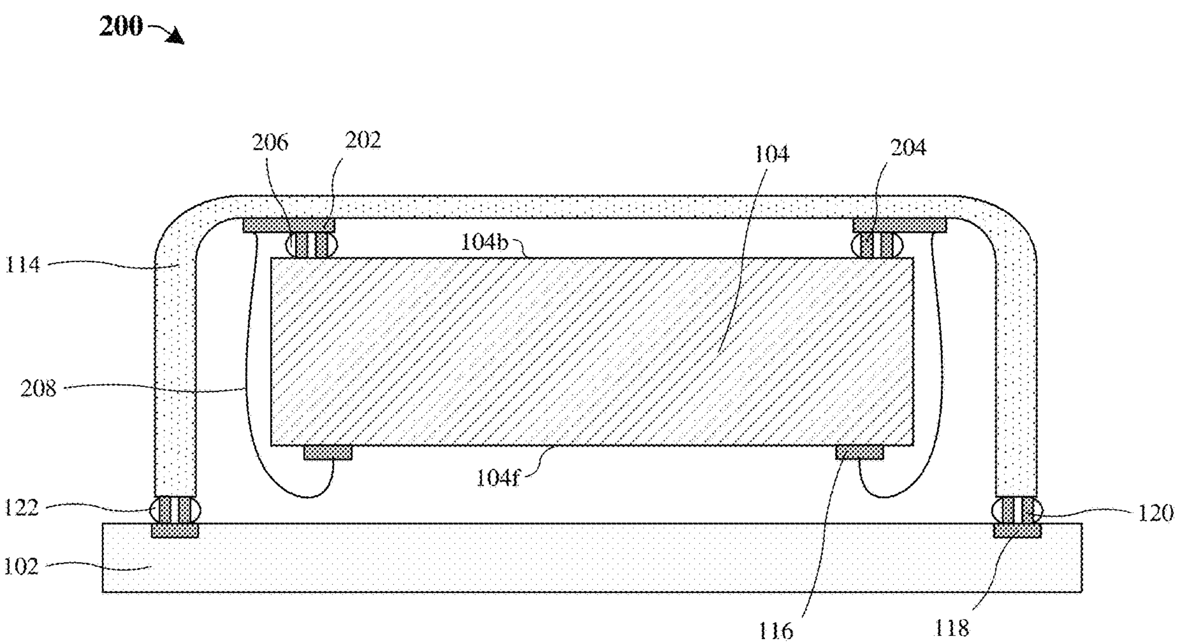
Fig. 1



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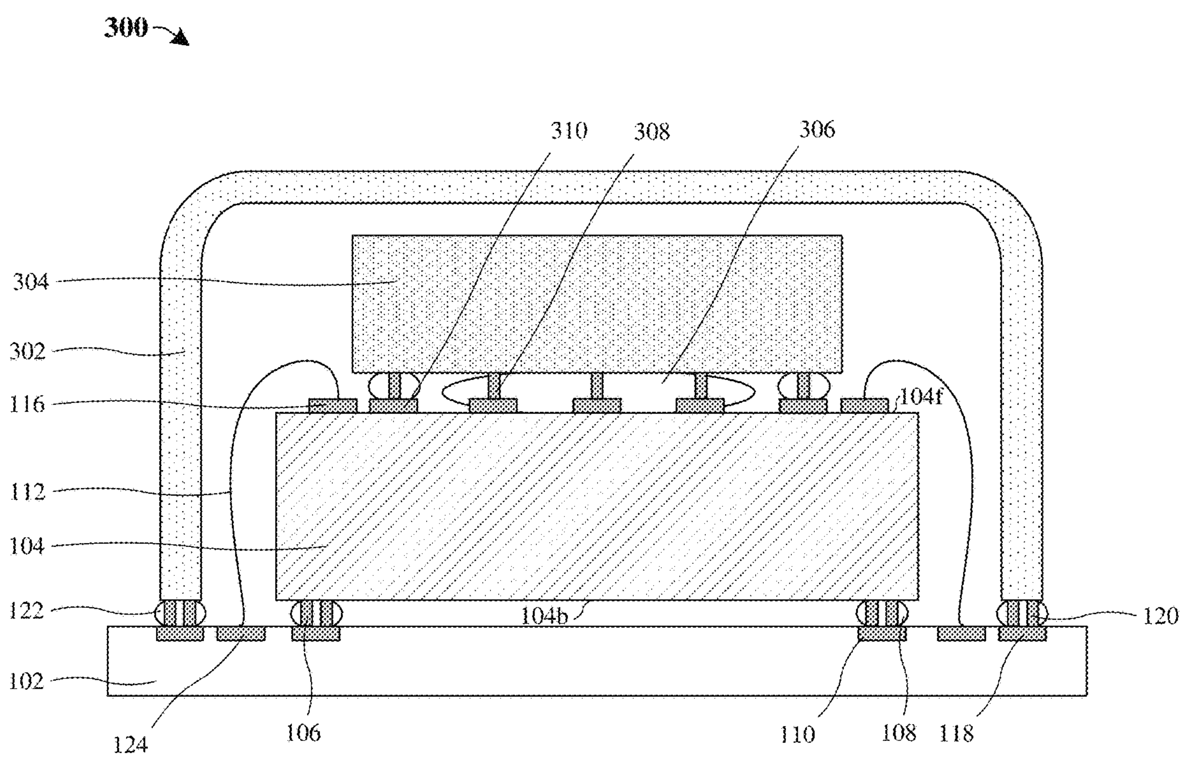
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116
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122
104b
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124
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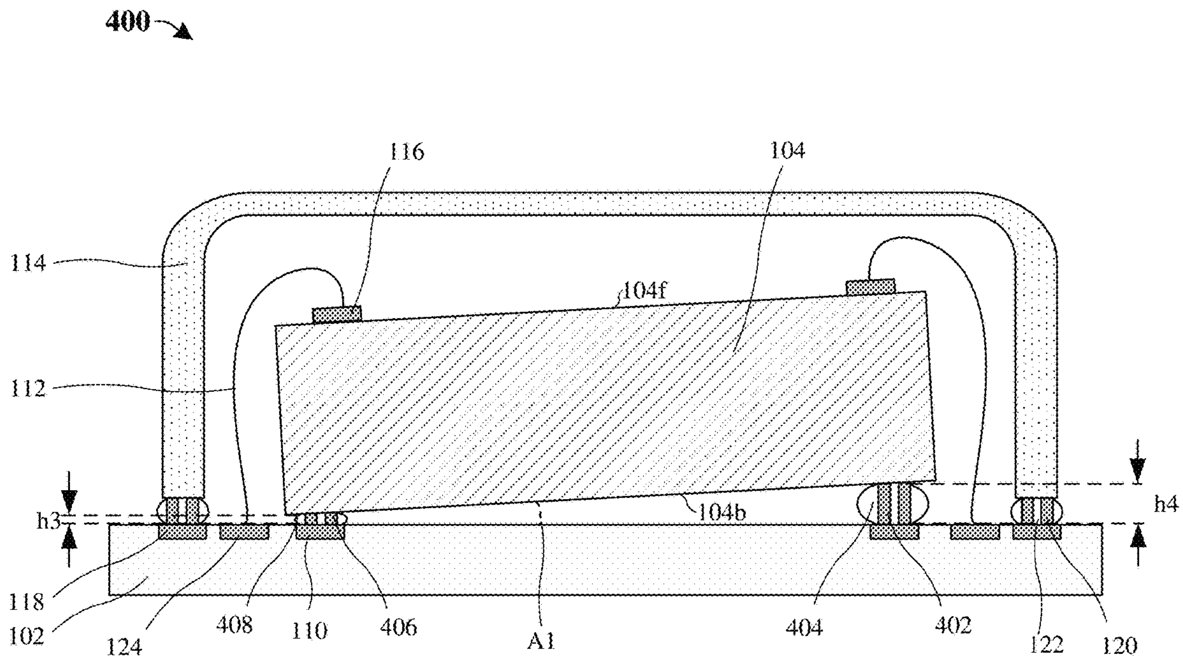
108
118

Fig. 3


400
116
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114
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h3
h4
104b
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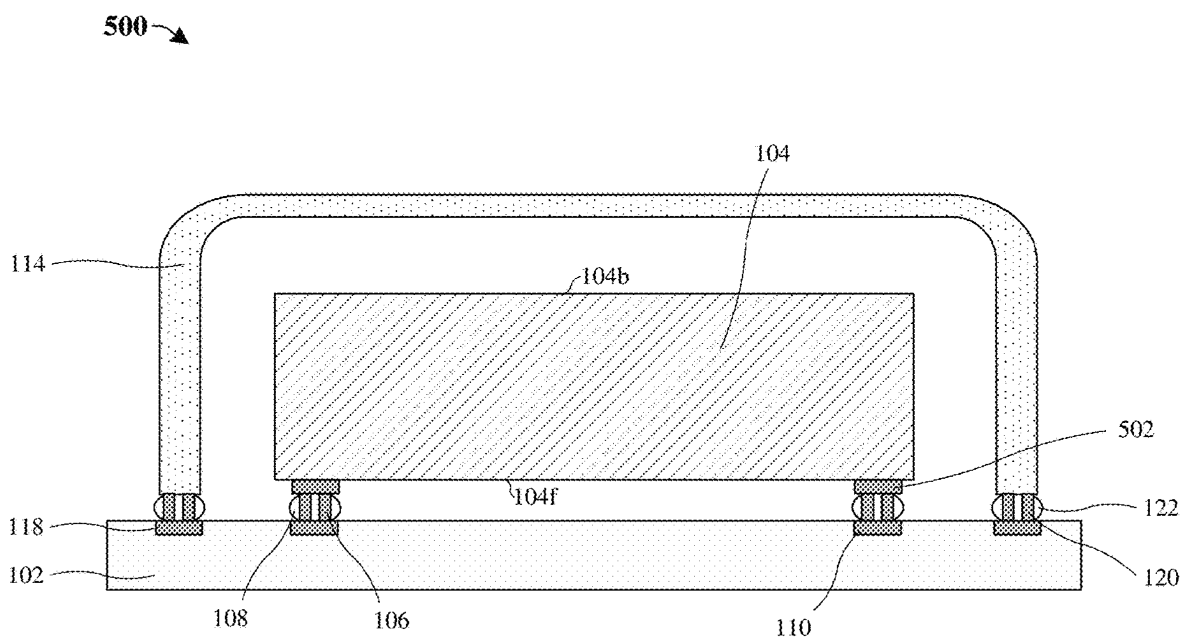
406
A1
404
402
122
120

Fig. 4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104f
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122
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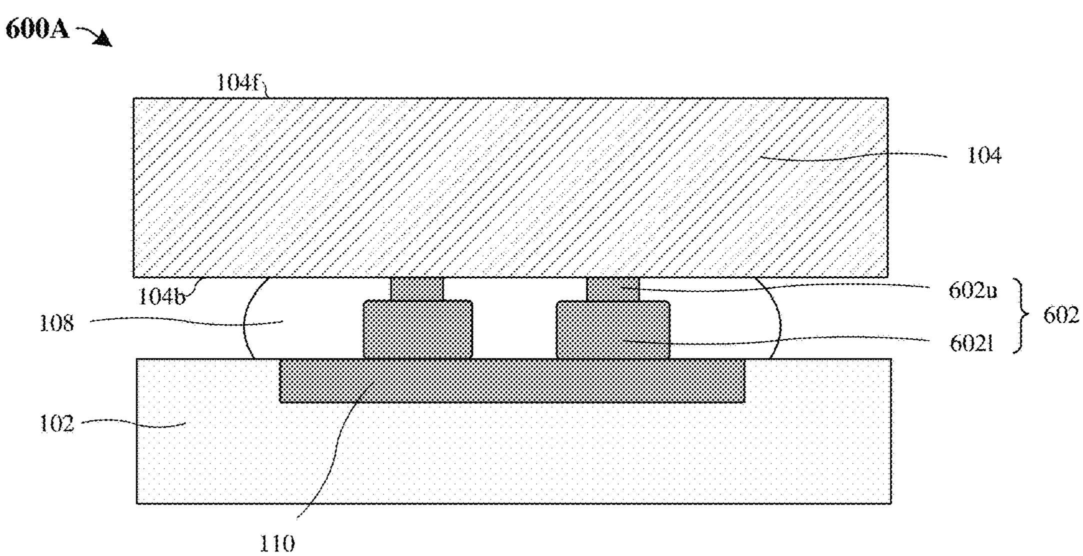
106
110

Fig. 5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108
602l
102
110

Fig. 6A

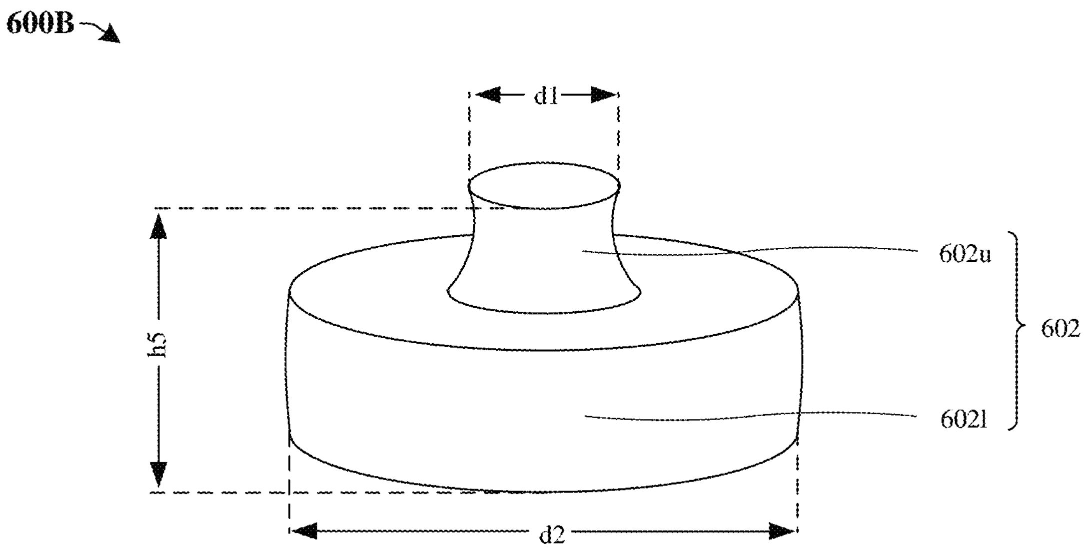


Fig. 6B

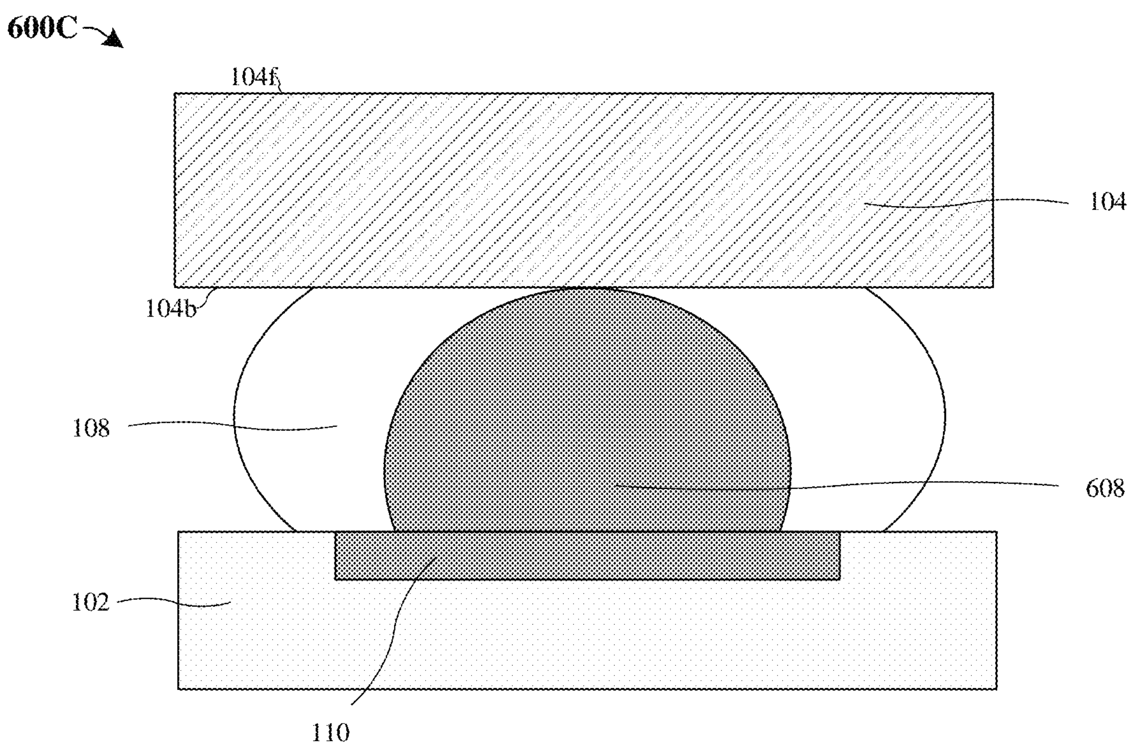


Fig. 6C

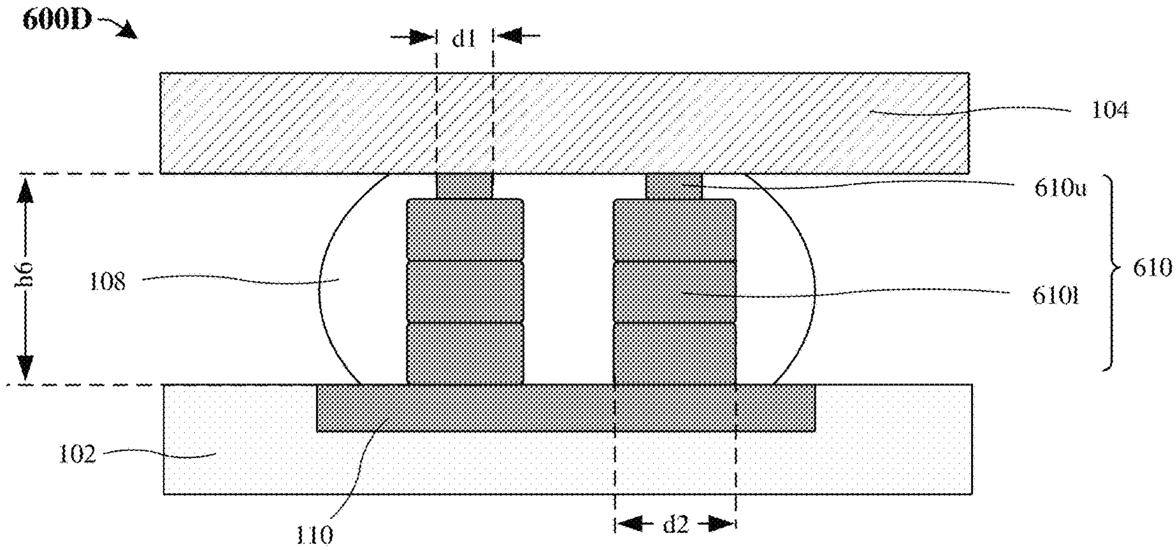


Fig. 6D

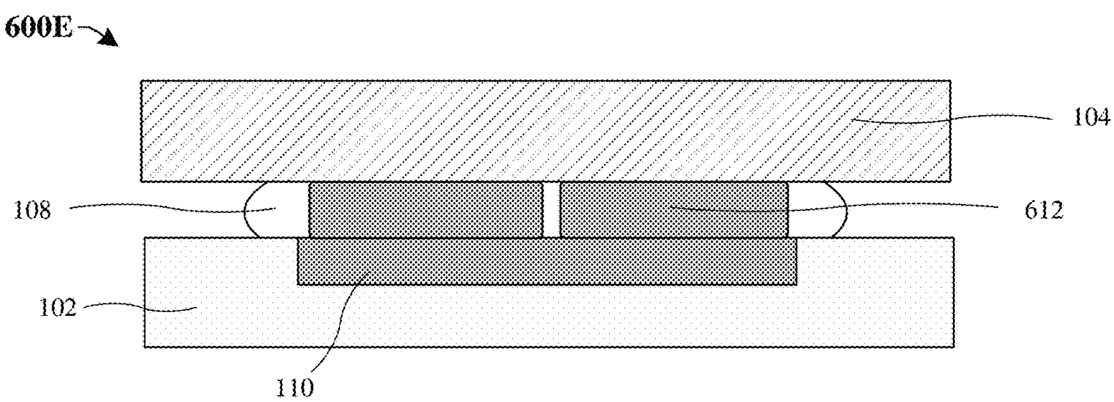
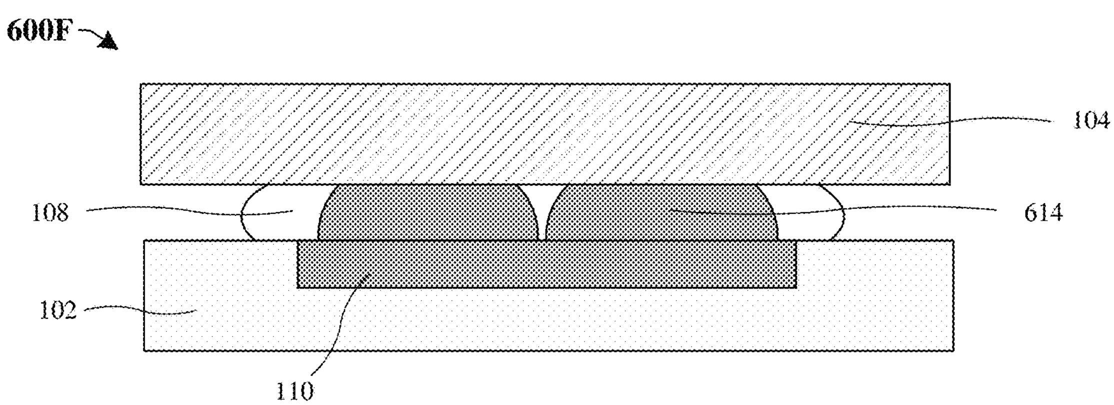


Fig. 6E



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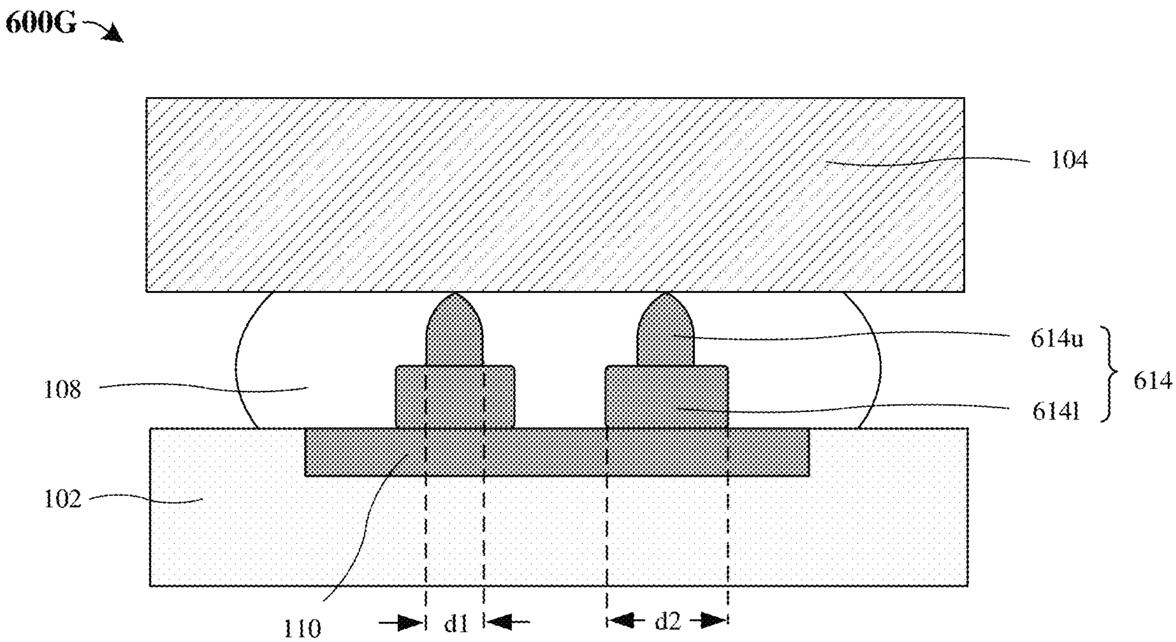
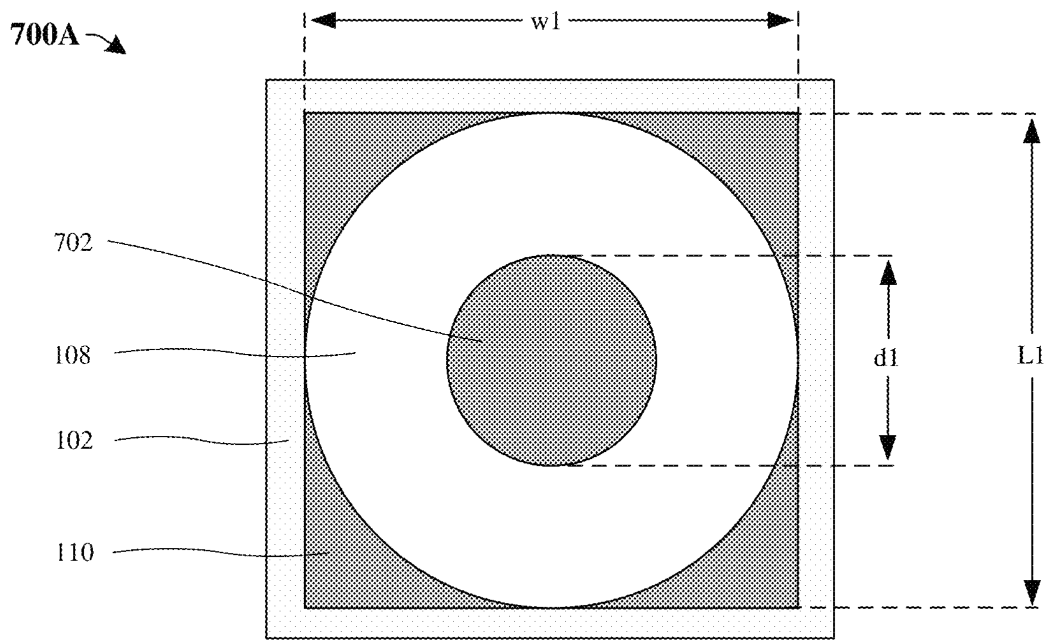


Fig. 6G



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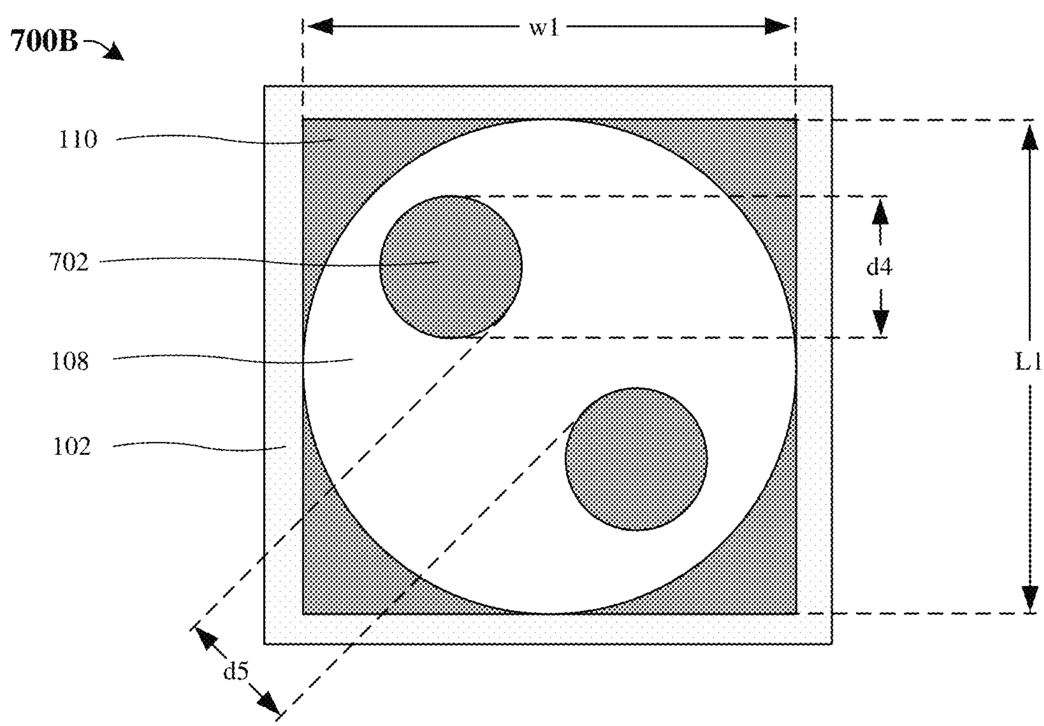


Fig. 7B

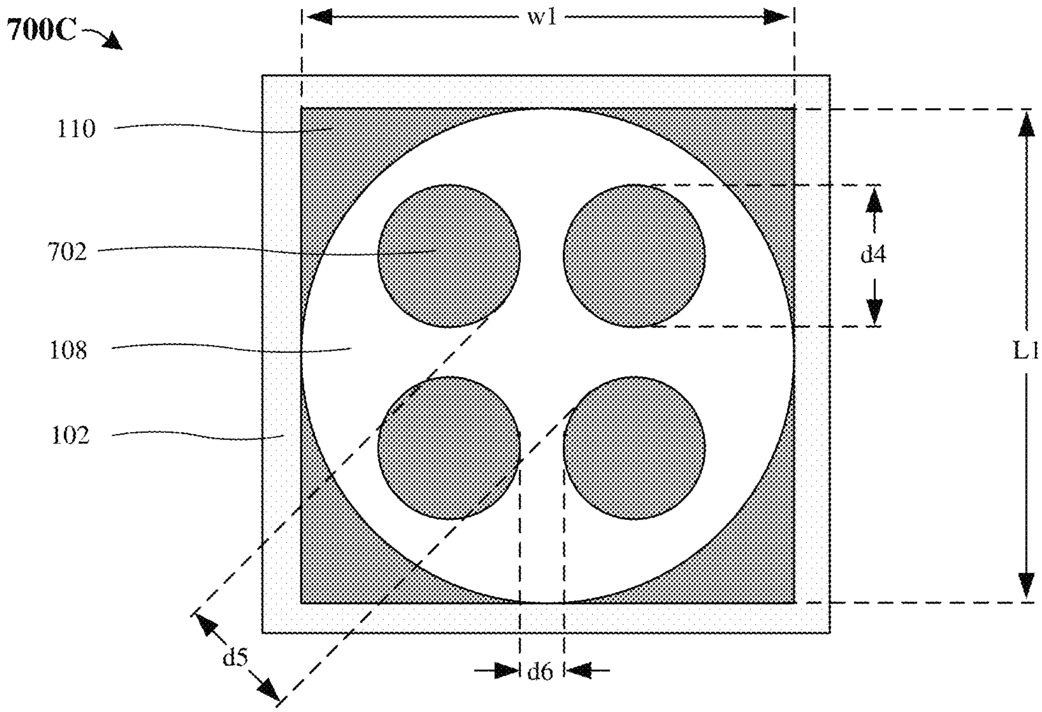


Fig. 7C

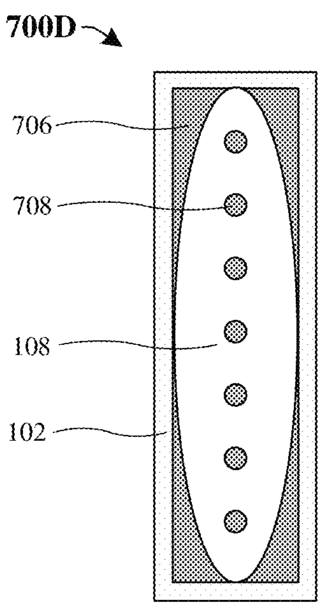


Fig. 7D

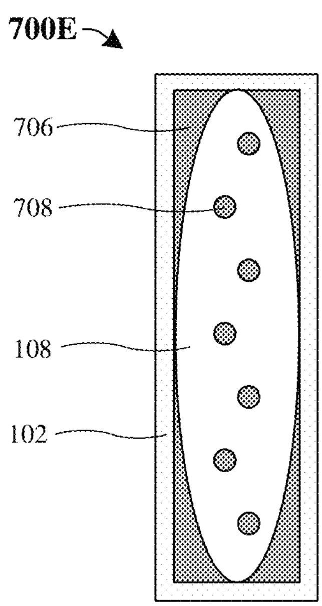


Fig. 7E

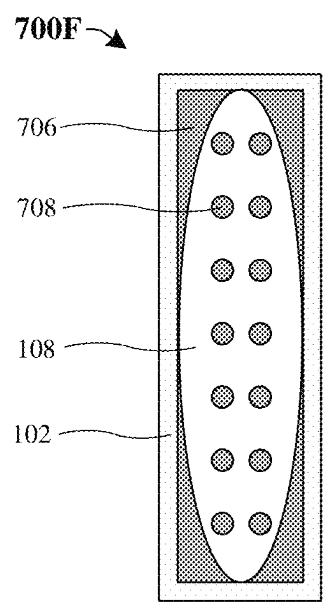


Fig. 7F

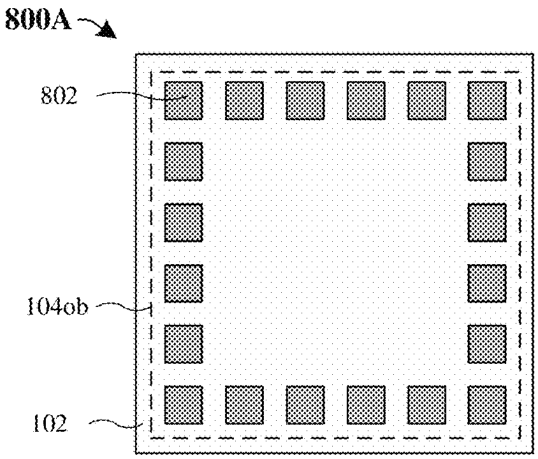
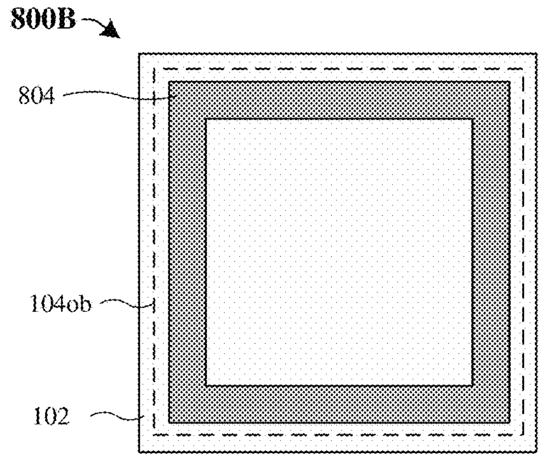


Fig. 8A



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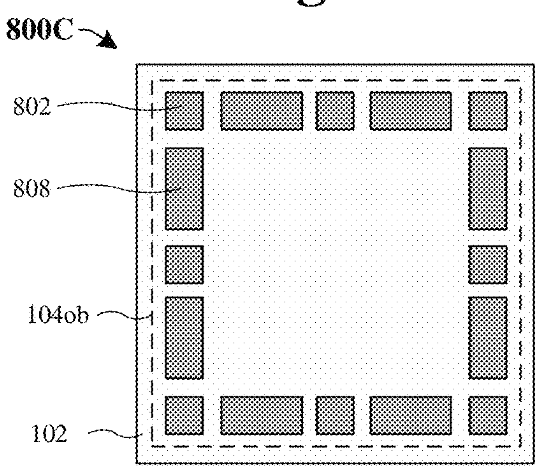

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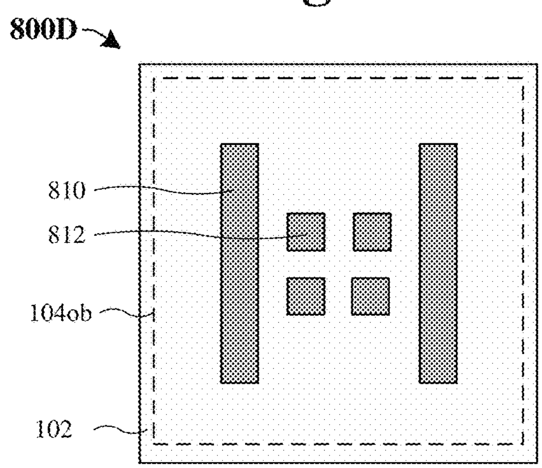

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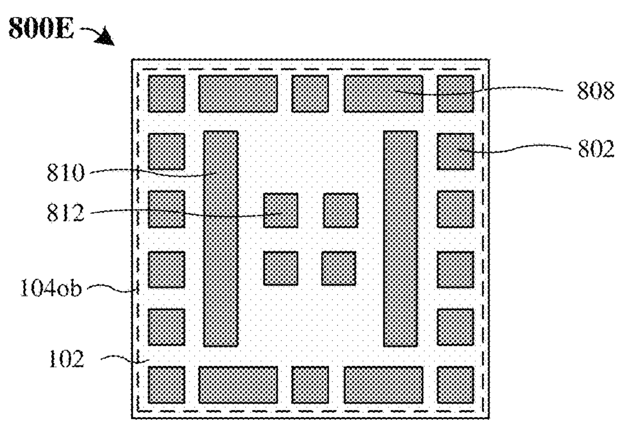

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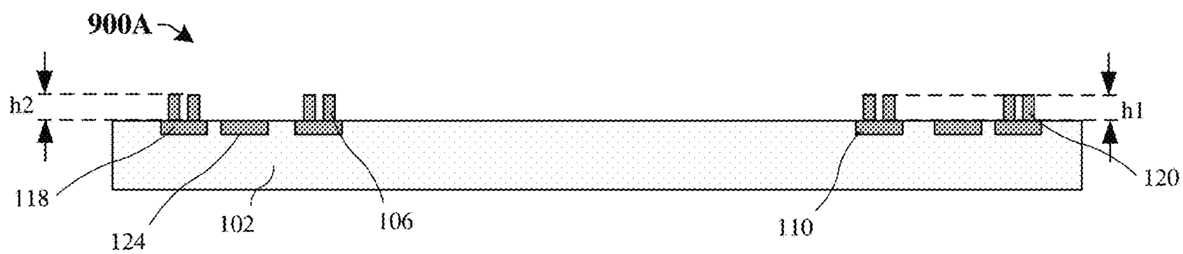


Fig. 9A

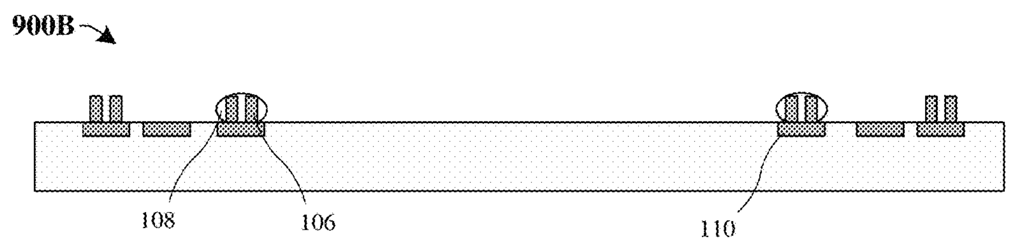
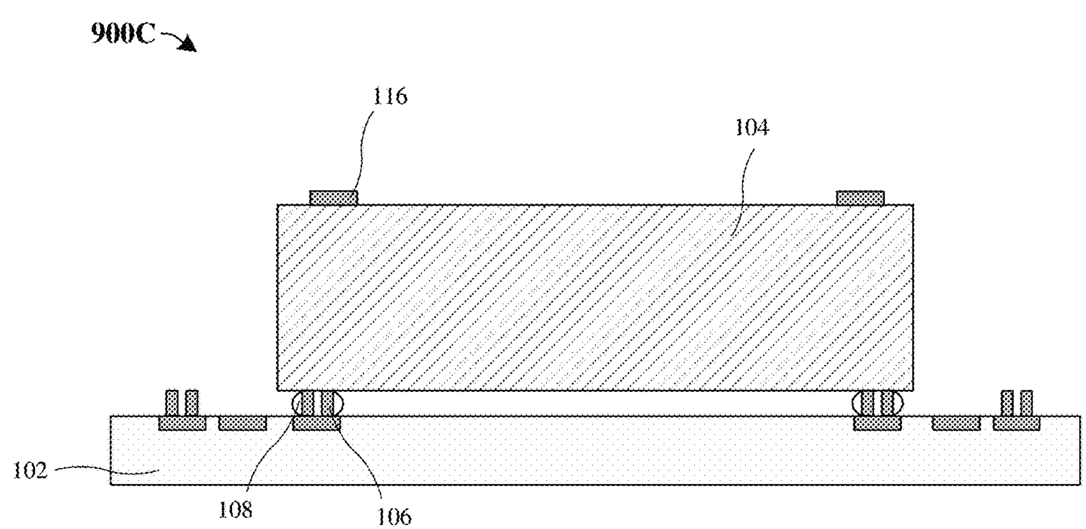


Fig. 9B



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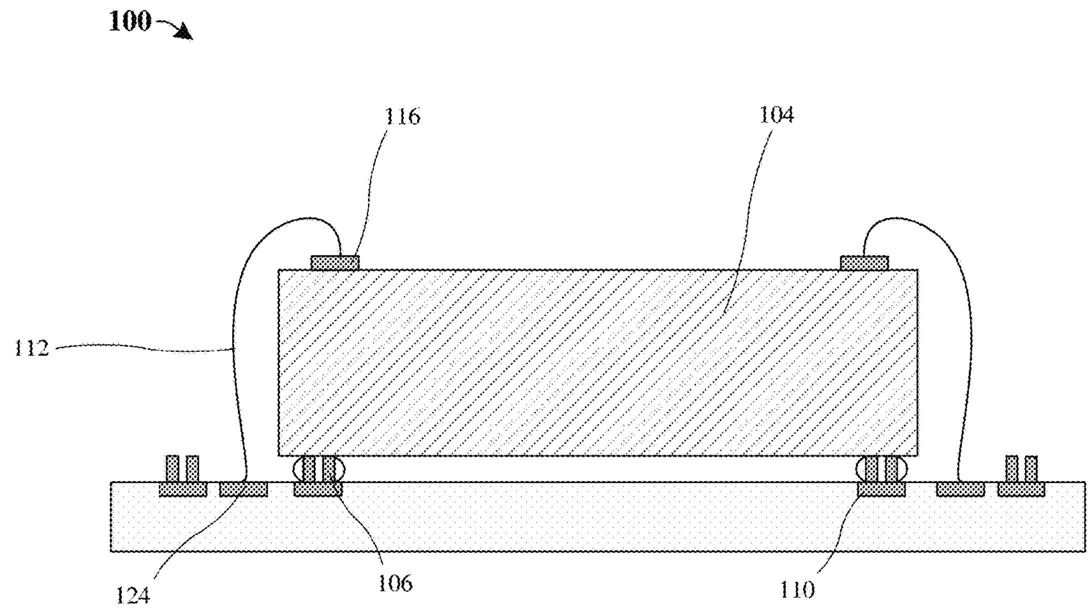

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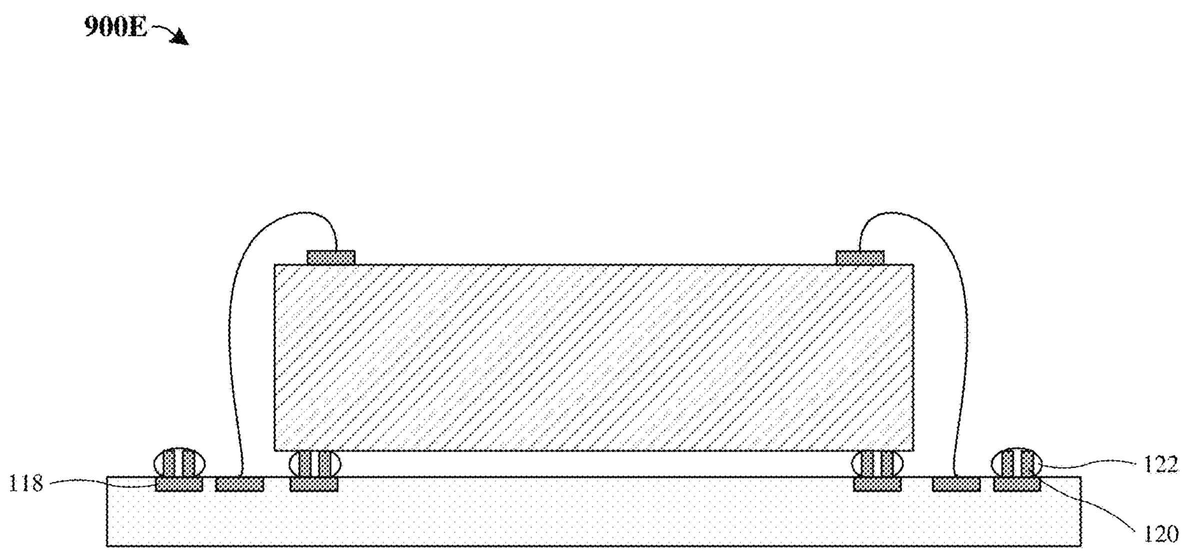

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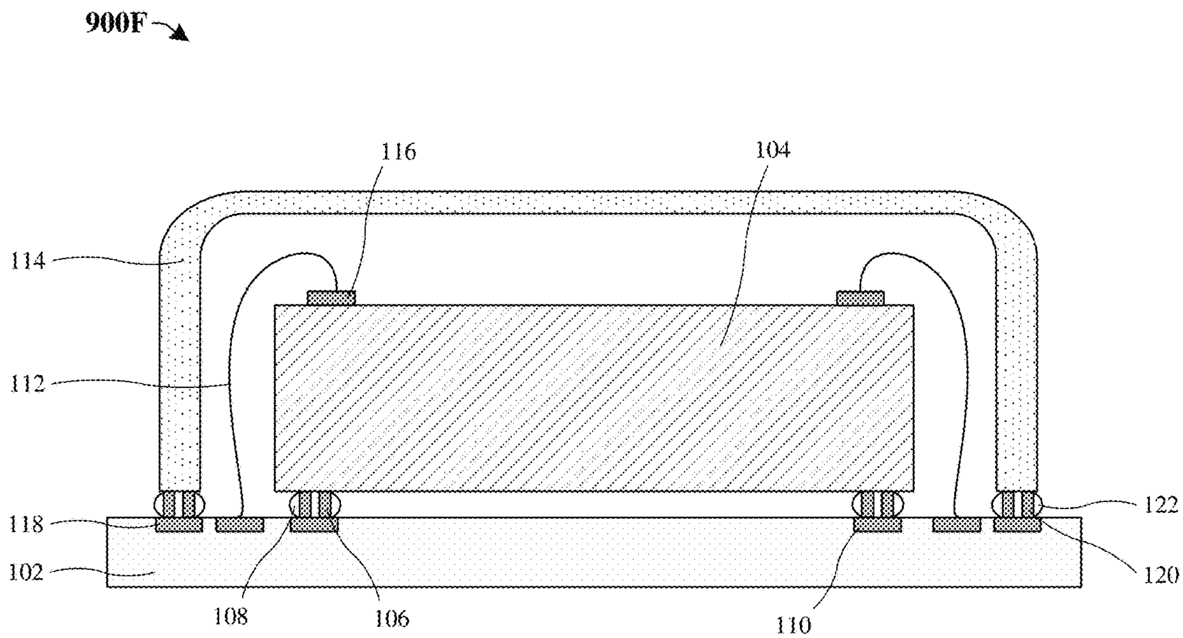

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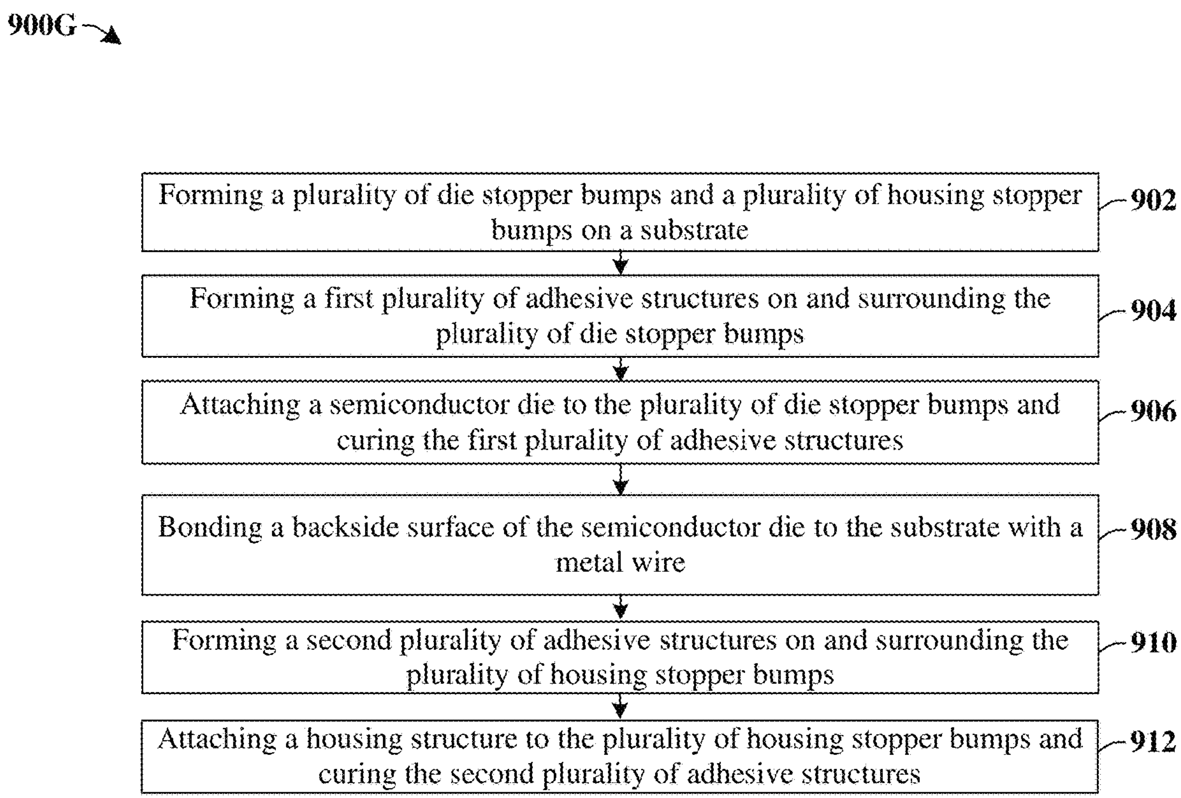

900G
Forming a plurality of die stopper bumps and a plurality of housing stopper bumps on a substrate
902
Forming a first plurality of adhesive structures on and surrounding the plurality of die stopper bumps
904
Attaching a semiconductor die to the plurality of die stopper bumps and curing the first plurality of adhesive structures
906
Bonding a backside surface of the semiconductor die to the substrate with a metal wire
908
Forming a second plurality of adhesive structures on and surrounding the plurality of housing stopper bumps
910
Attaching a housing structure to the plurality of housing stopper bumps and curing the second plurality of adhesive structures
912

Fig. 9G

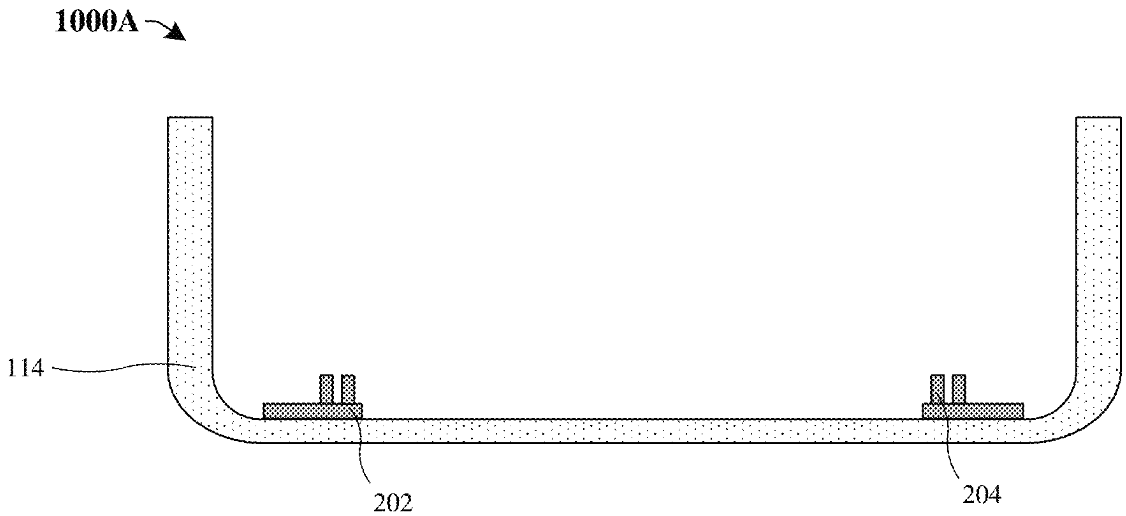


Fig. 10A

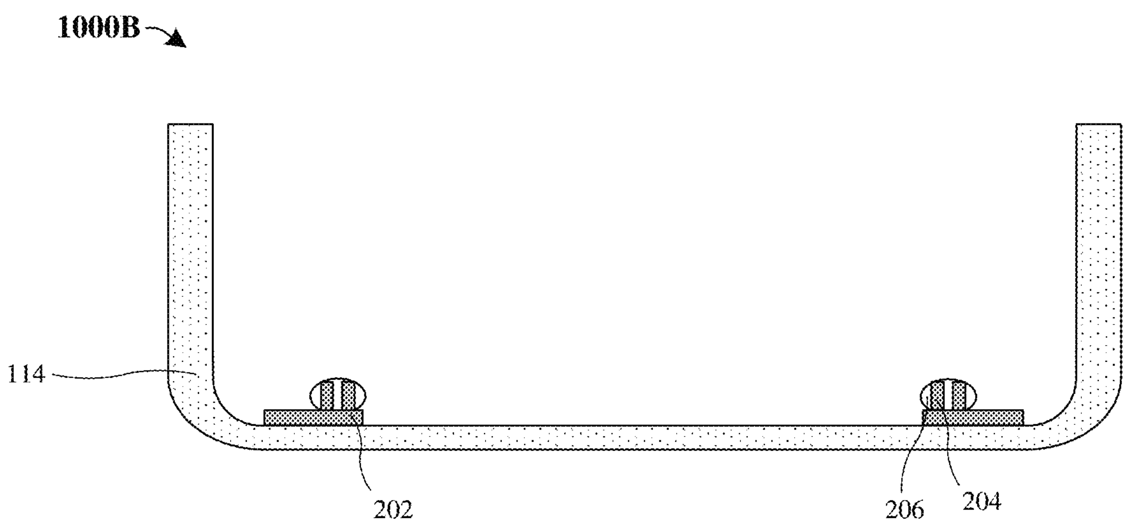


Fig. 10B

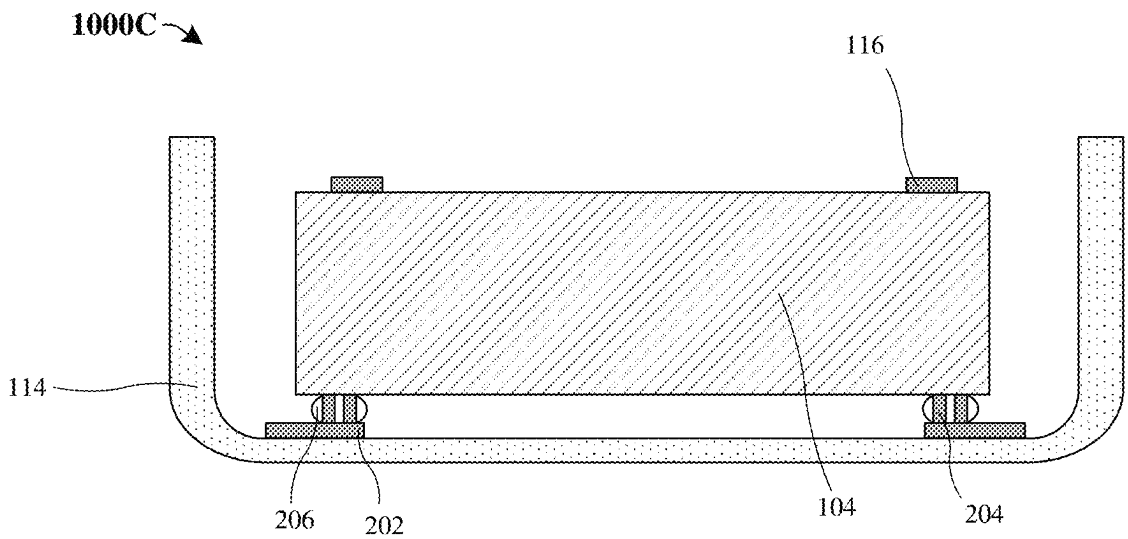


Fig. 10C

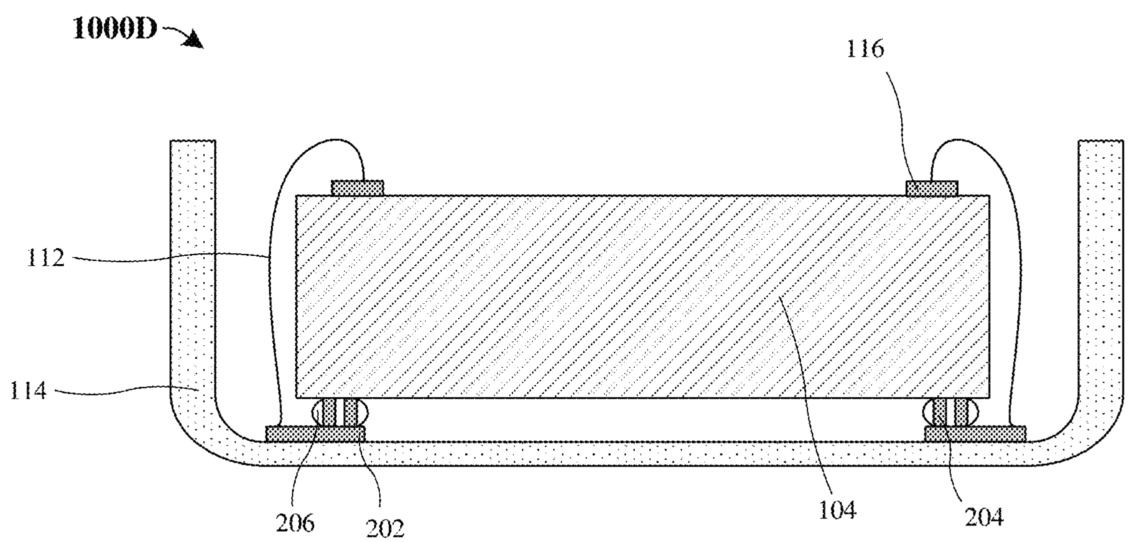


Fig. 10D

Fig. 10E

Fig. 10F

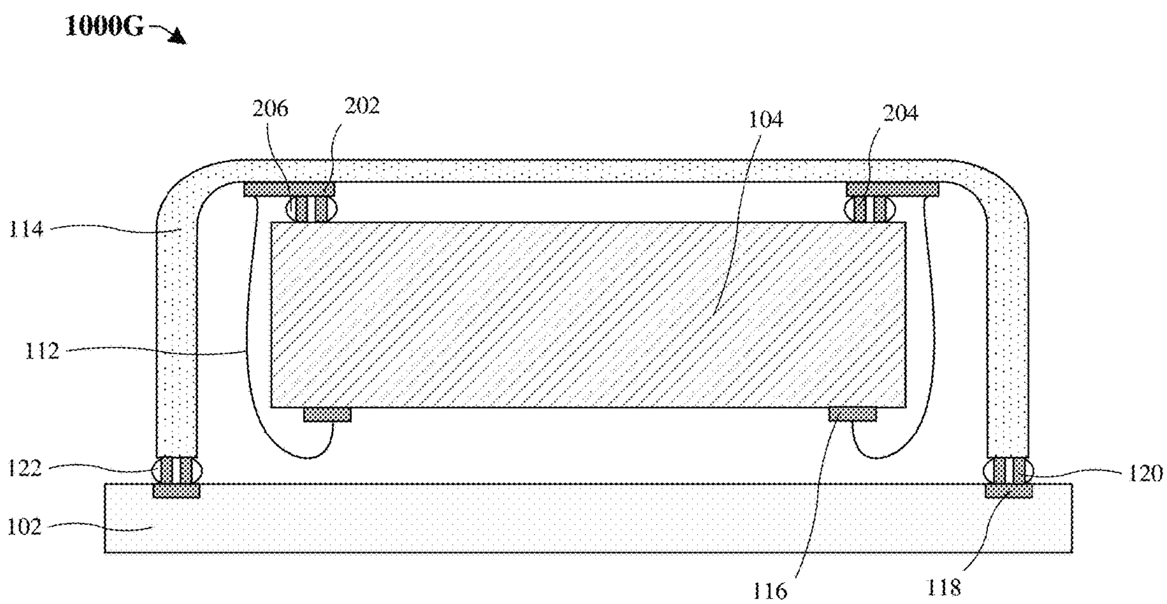


Fig. 10G

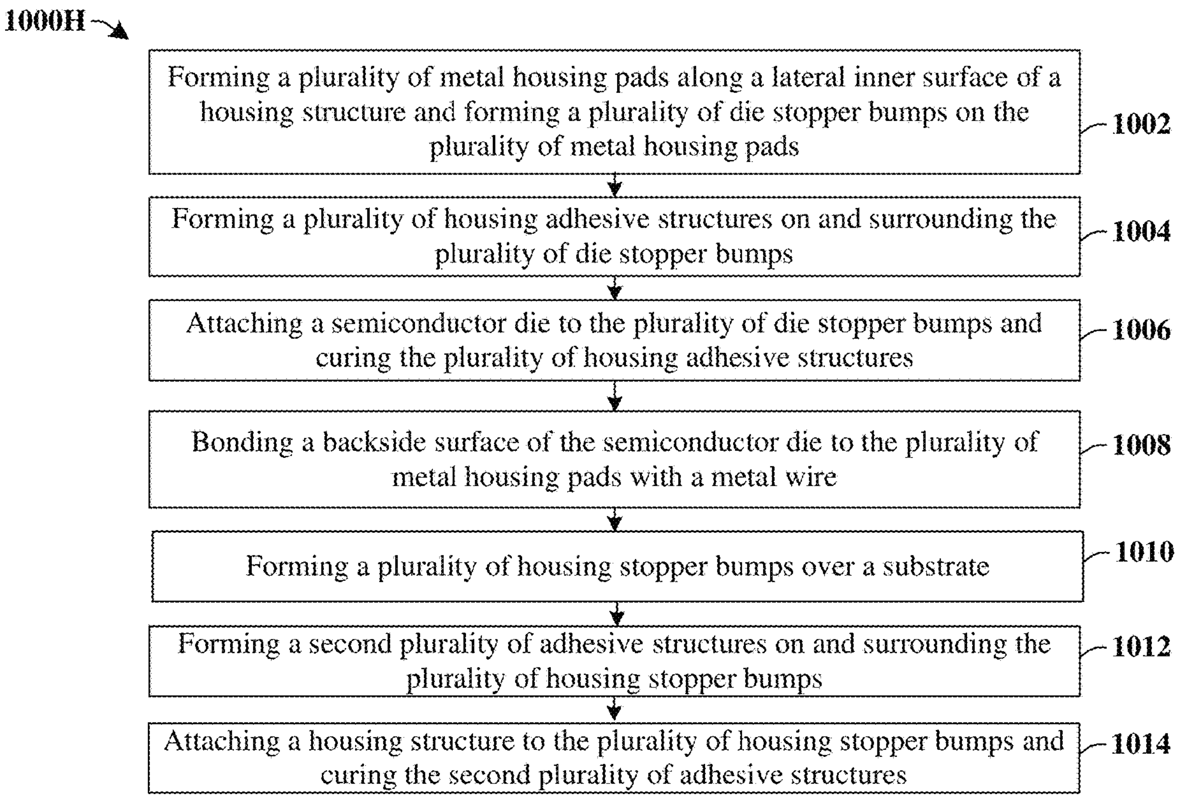

1000H
Forming a plurality of metal housing pads along a lateral inner surface of a housing structure and forming a plurality of die stopper bumps on the plurality of metal housing pads
1002
Forming a plurality of housing adhesive structures on and surrounding the plurality of die stopper bumps
1004
Attaching a semiconductor die to the plurality of die stopper bumps and curing the plurality of housing adhesive structures
1006
Bonding a backside surface of the semiconductor die to the plurality of metal housing pads with a metal wire
1008
Forming a plurality of housing stopper bumps over a substrate
1010
Forming a second plurality of adhesive structures on and surrounding the plurality of housing stopper bumps
1012
Attaching a housing structure to the plurality of housing stopper bumps and curing the second plurality of adhesive structures
1014

Fig. 10H

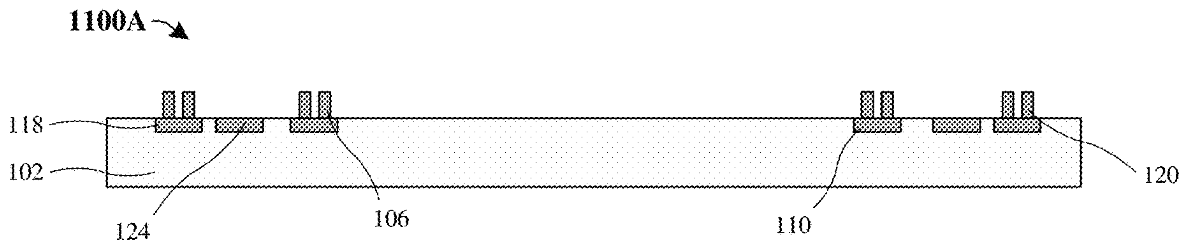


Fig. 11A

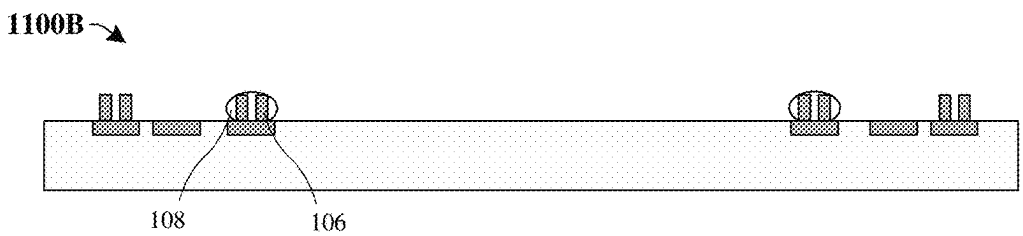


Fig. 11B

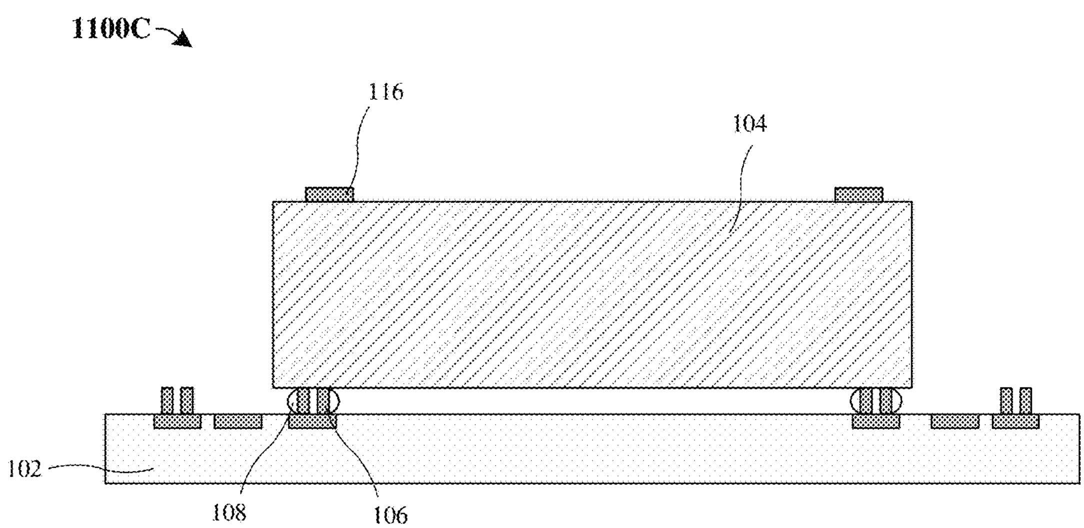
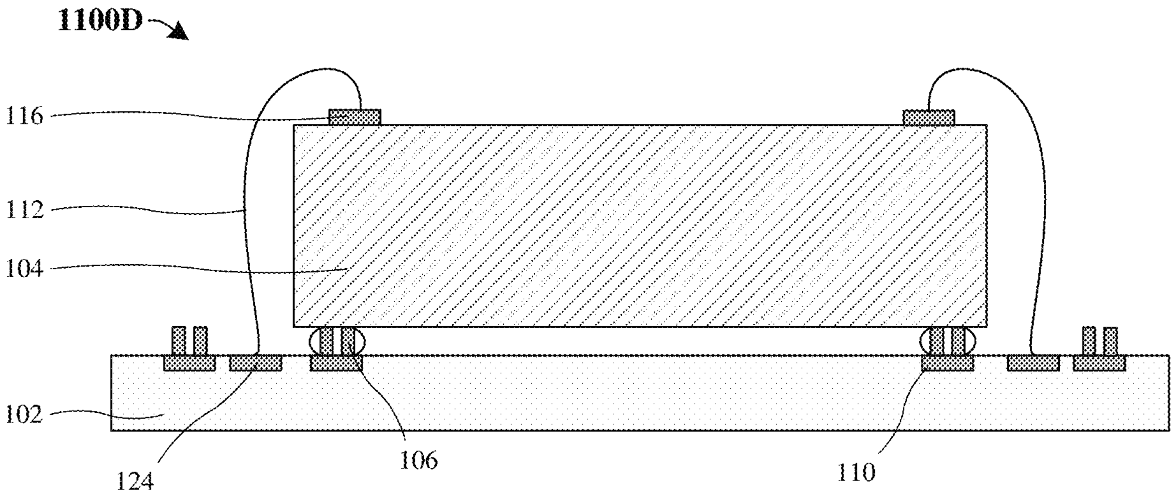


Fig. 11C



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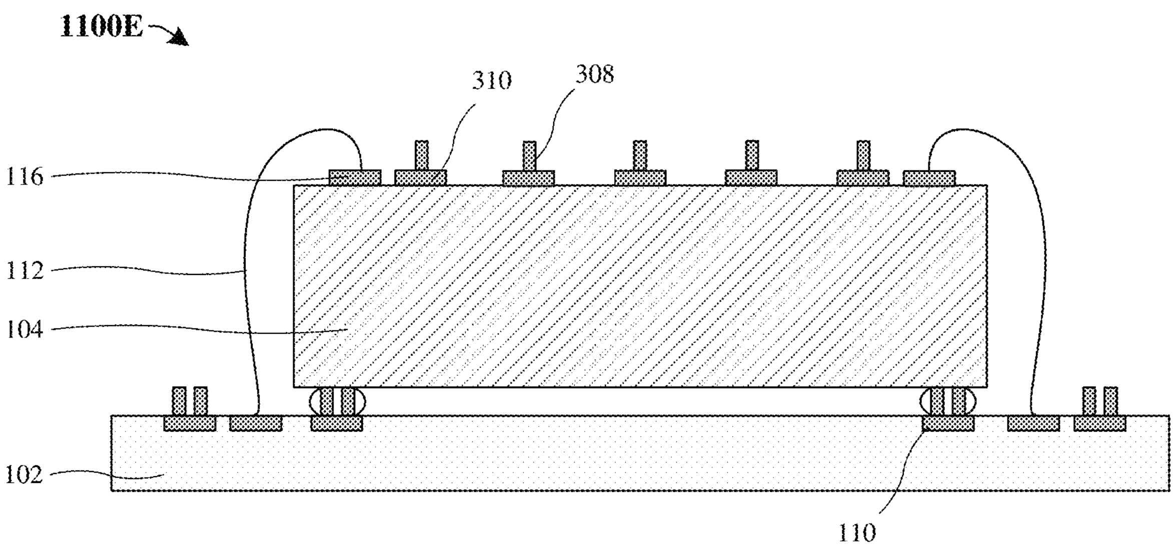

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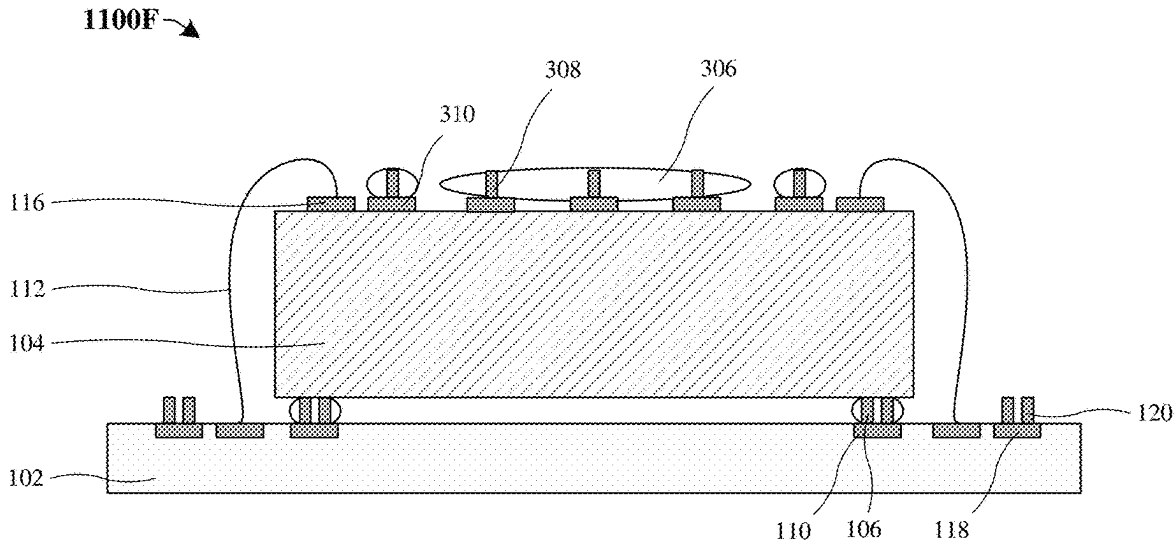

Fig. 11F

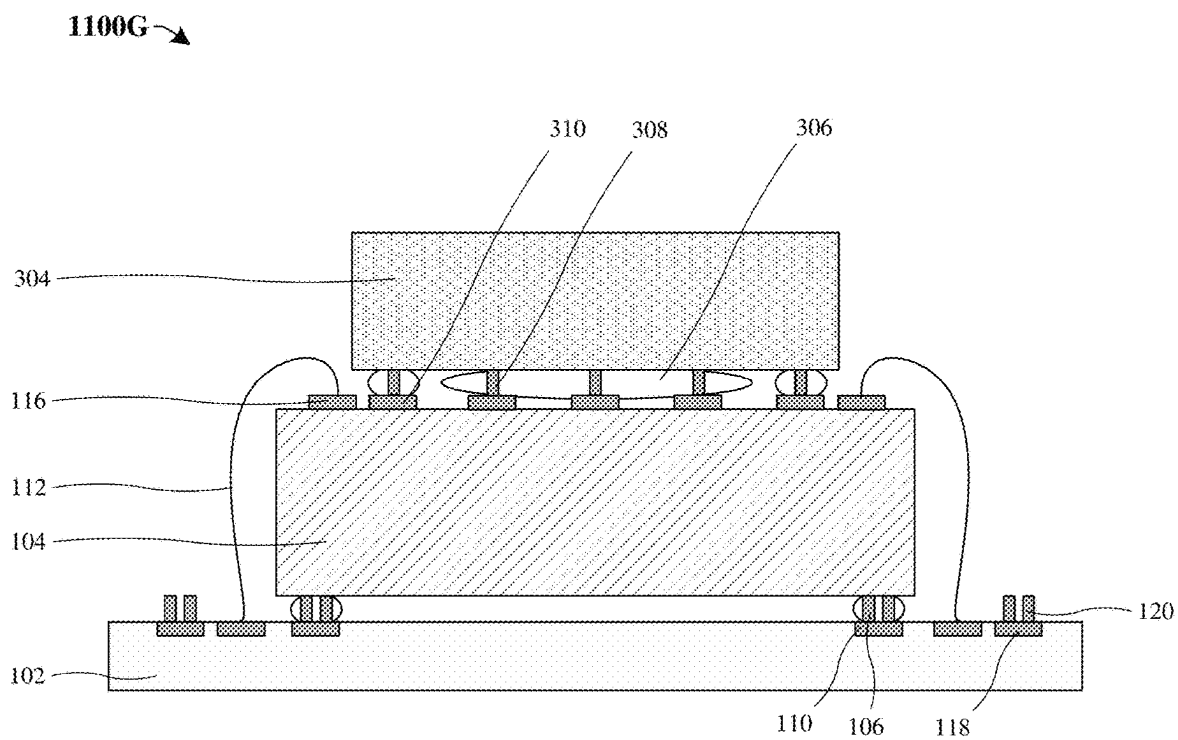

Fig. 11G

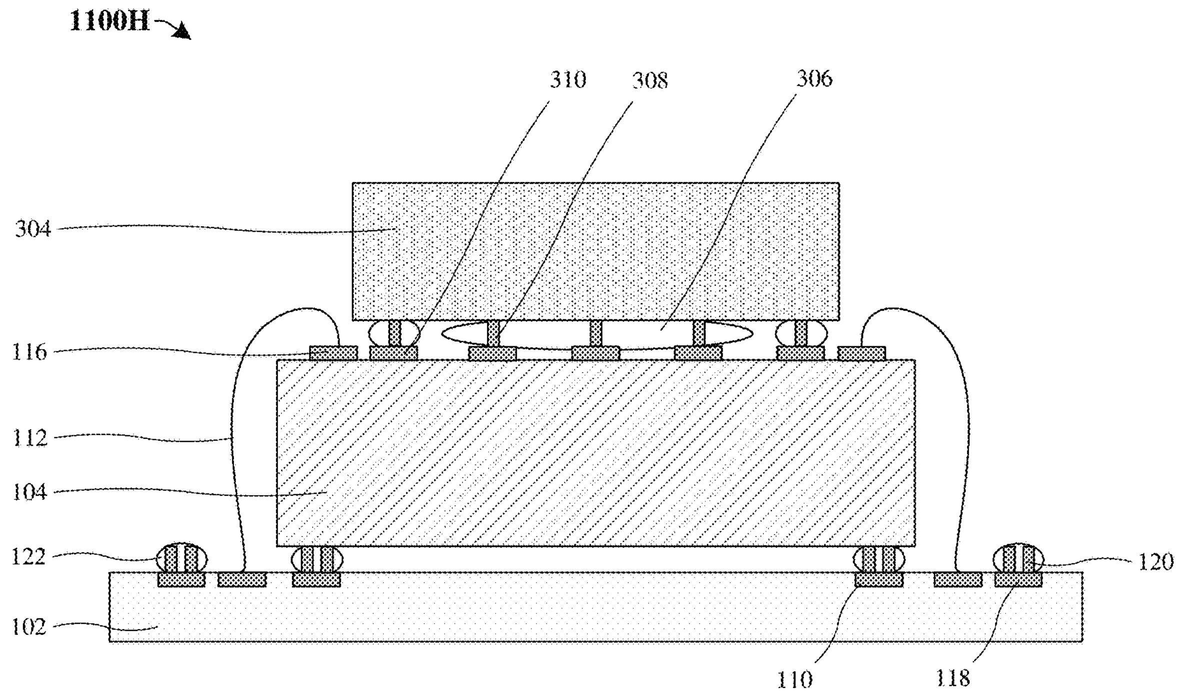

Fig. 11H

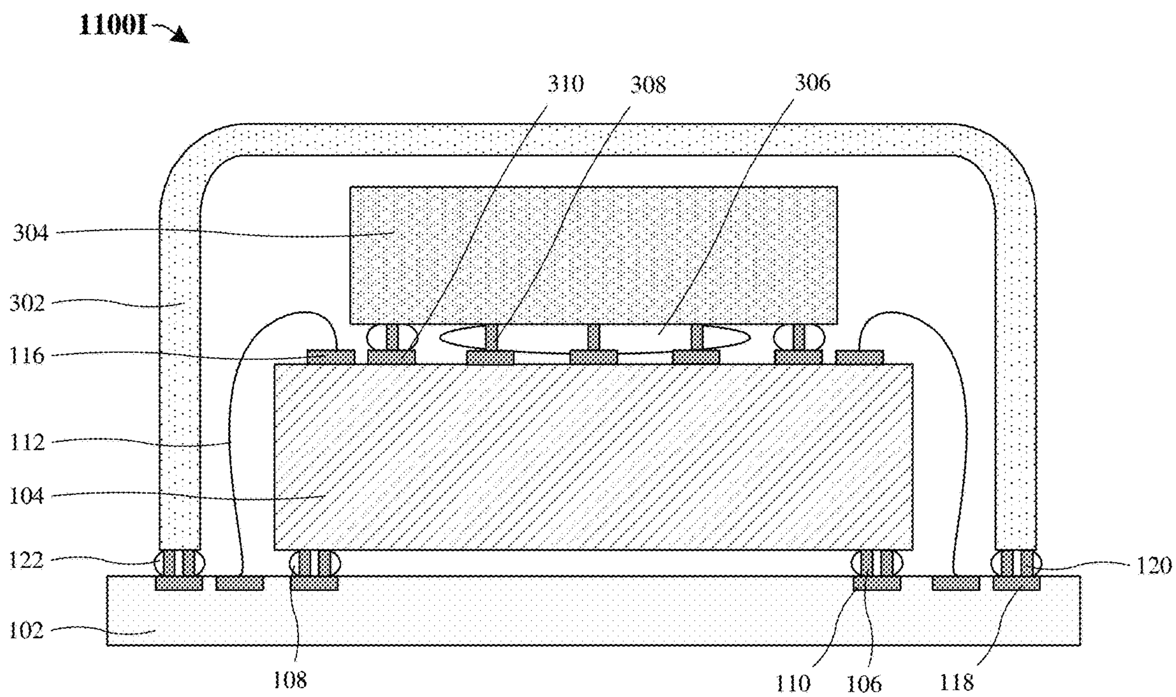

Fig. 11I

1100J

Forming a plurality of die stopper bumps and a plurality of housing stopper bumps on a substrate 1102

↓

Forming a first plurality of adhesive structures on and surrounding the plurality of die stopper bumps 1104

↓

Attaching a semiconductor die to the plurality of die stopper bumps and curing the first plurality of adhesive structures 1106

↓

Bonding a backside surface of the semiconductor die to the substrate with a metal wire 1108

↓

Forming a plurality of metal die pads along a backside surface of the semiconductor die, and forming a plurality of overlying die stopper bumps over the plurality of metal die pads 1110

↓

Forming a plurality of overlying adhesive structures on and surrounding each of the plurality of overlying die stopper bumps 1112

↓

Attaching a stacked semiconductor die to the plurality of overlying die stopper bumps and curing the plurality of overlying adhesive structures 1114

↓

Forming a second plurality of adhesive structures on and surrounding the plurality of housing stopper bumps 1116

↓

Attaching a housing structure to the plurality of housing stopper bumps and curing the second plurality of adhesive structures 1118

Fig. 11J

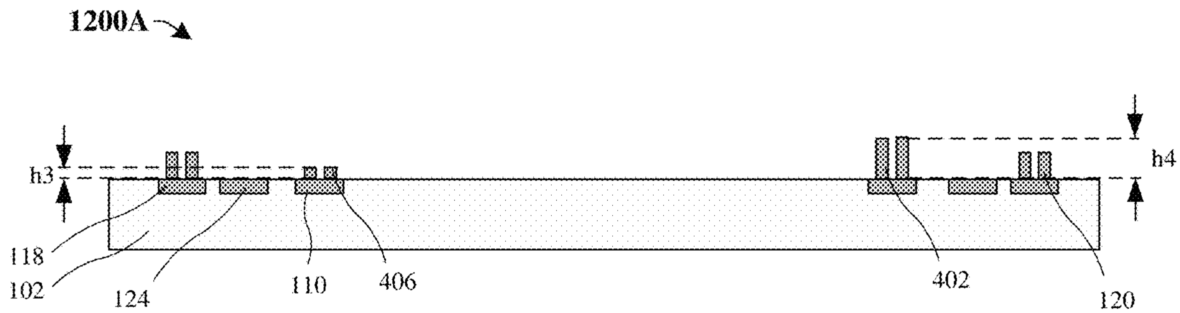

Fig. 12A

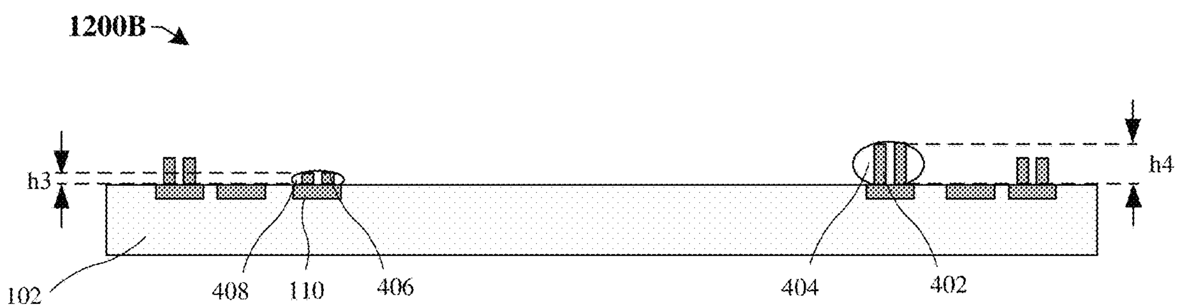

Fig. 12B

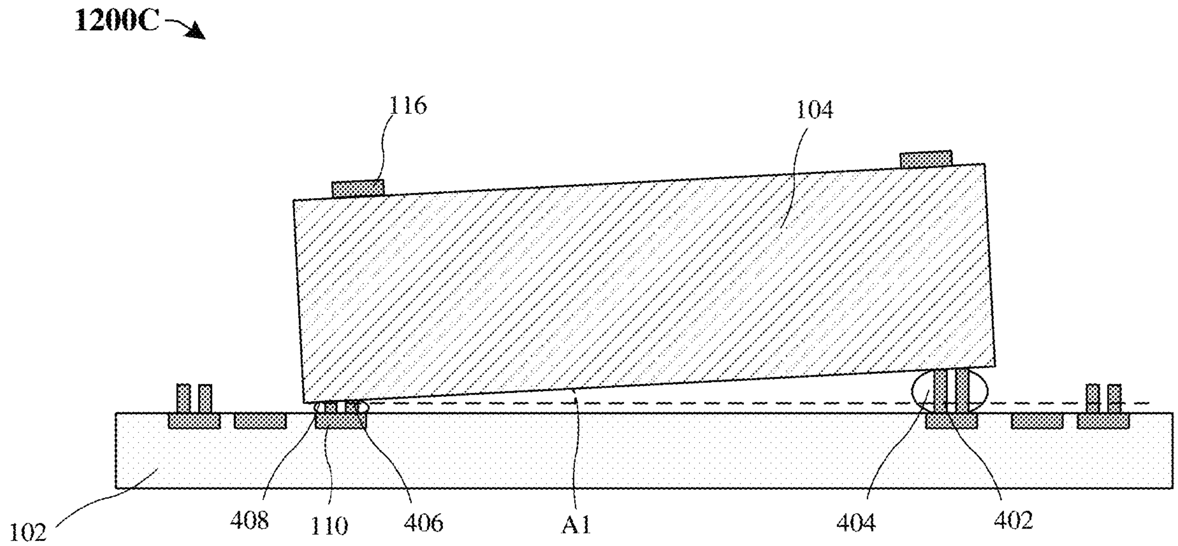

Fig. 12C

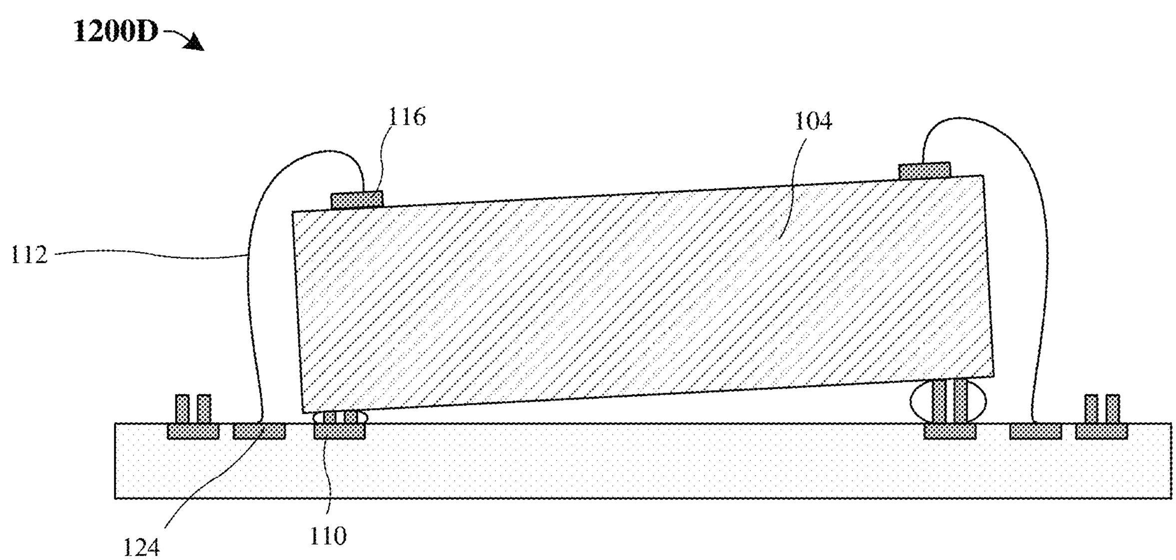

Fig. 12D

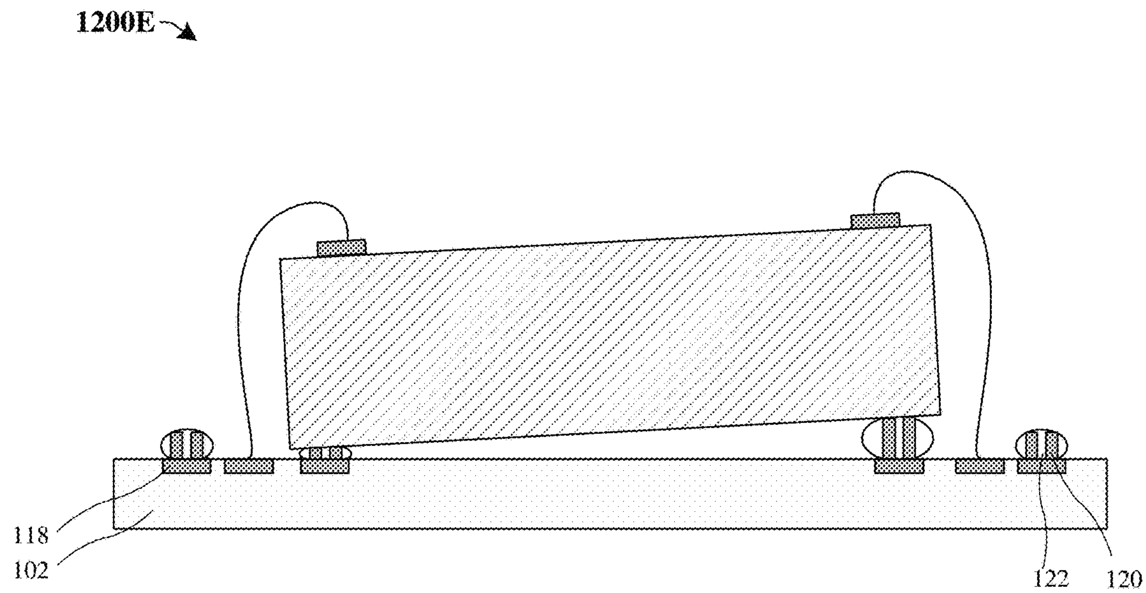

Fig. 12E

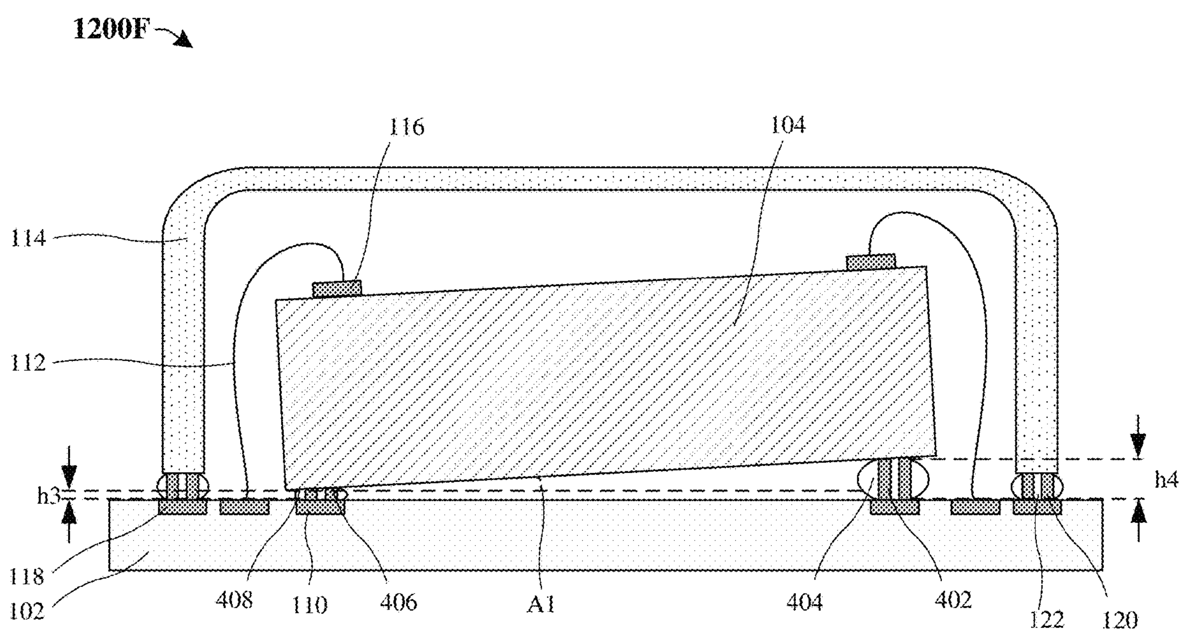

Fig. 12F

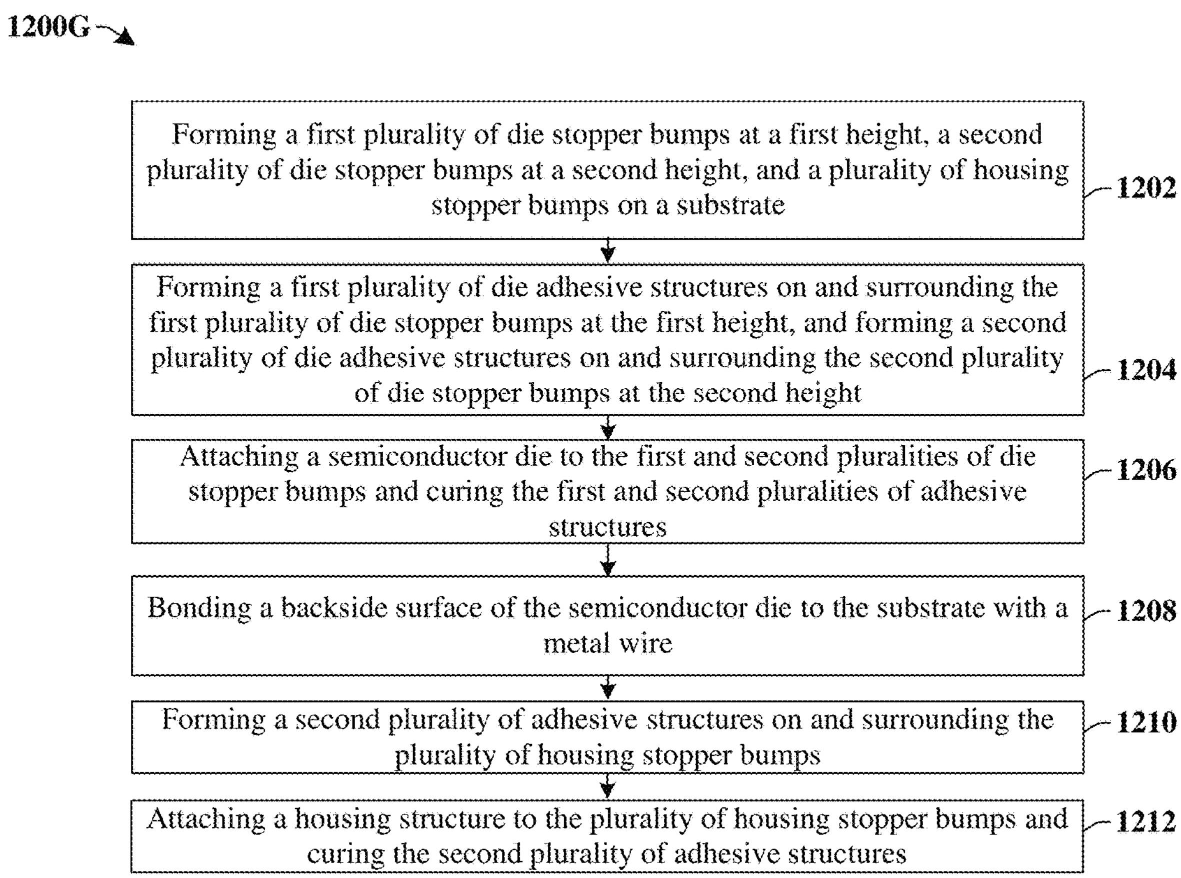
1200G
Forming a first plurality of die stopper bumps at a first height, a second plurality of die stopper bumps at a second height, and a plurality of housing stopper bumps on a substrate
1202
Forming a first plurality of die adhesive structures on and surrounding the first plurality of die stopper bumps at the first height, and forming a second plurality of die adhesive structures on and surrounding the second plurality of die stopper bumps at the second height
1204
Attaching a semiconductor die to the first and second pluralities of die stopper bumps and curing the first and second pluralities of adhesive structures
1206
Bonding a backside surface of the semiconductor die to the substrate with a metal wire
1208
Forming a second plurality of adhesive structures on and surrounding the plurality of housing stopper bumps
1210
Attaching a housing structure to the plurality of housing stopper bumps and curing the second plurality of adhesive structures
1212

Fig. 12G

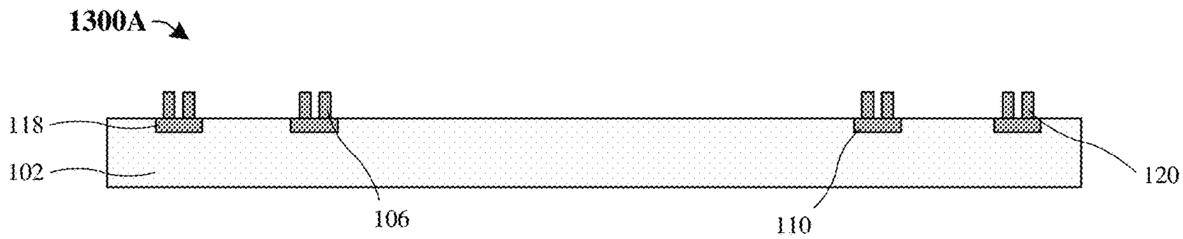

Fig. 13A

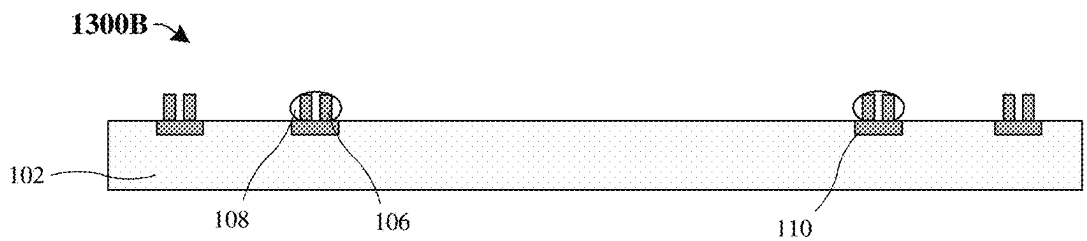

Fig. 13B

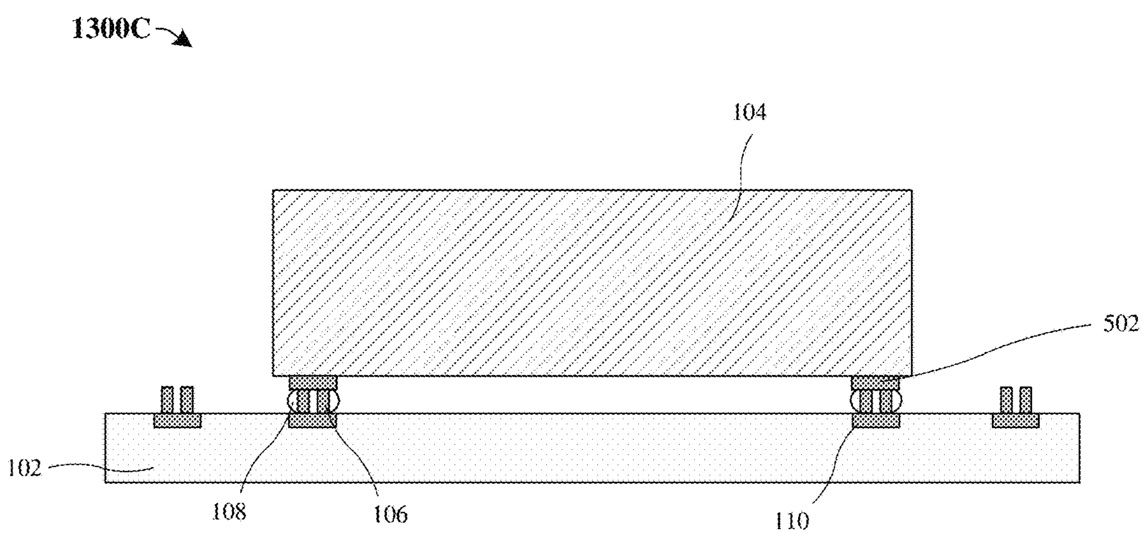

Fig. 13C

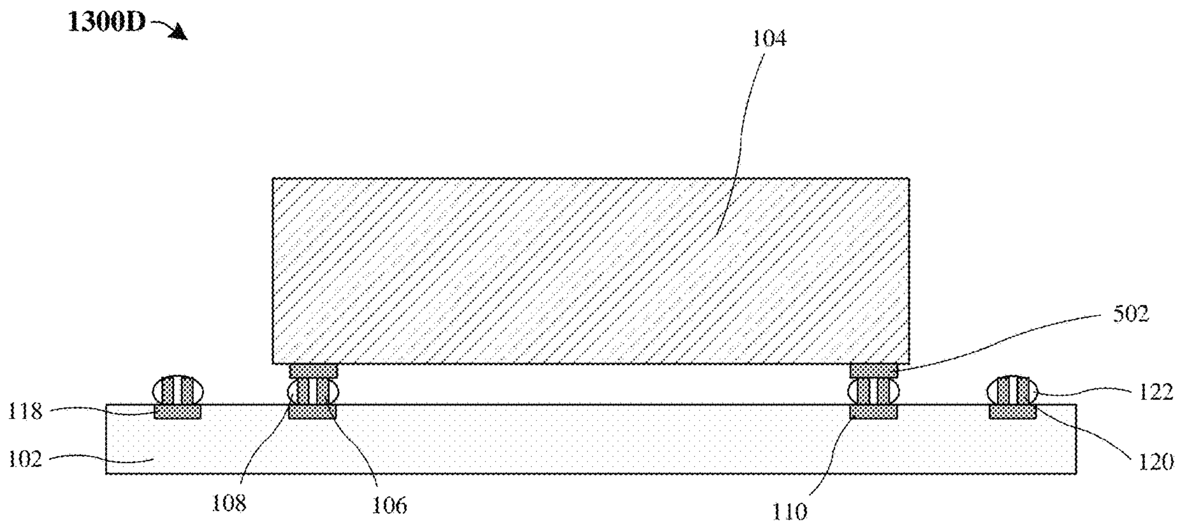

Fig. 13D

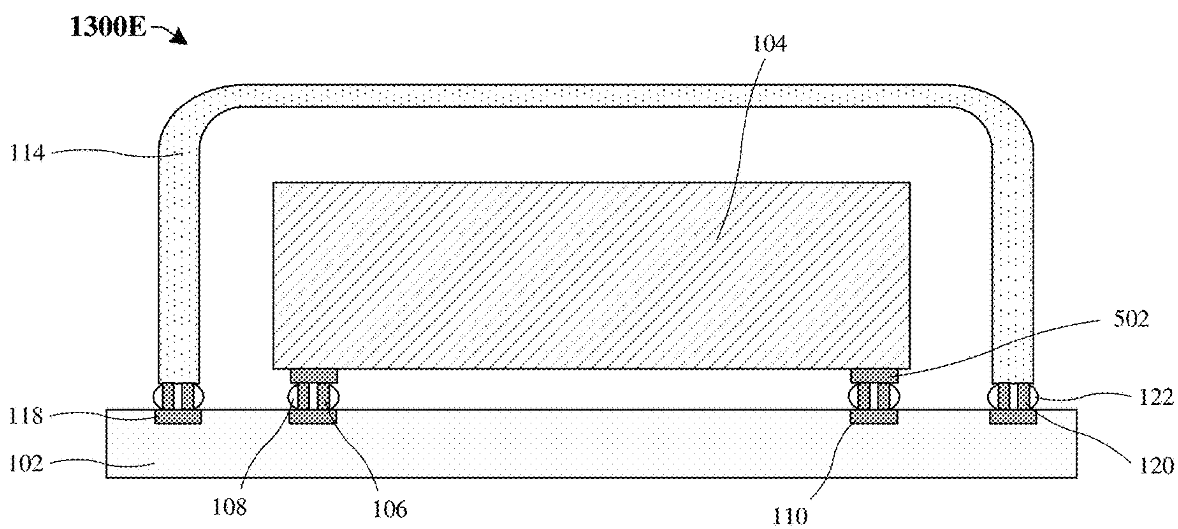

Fig. 13E

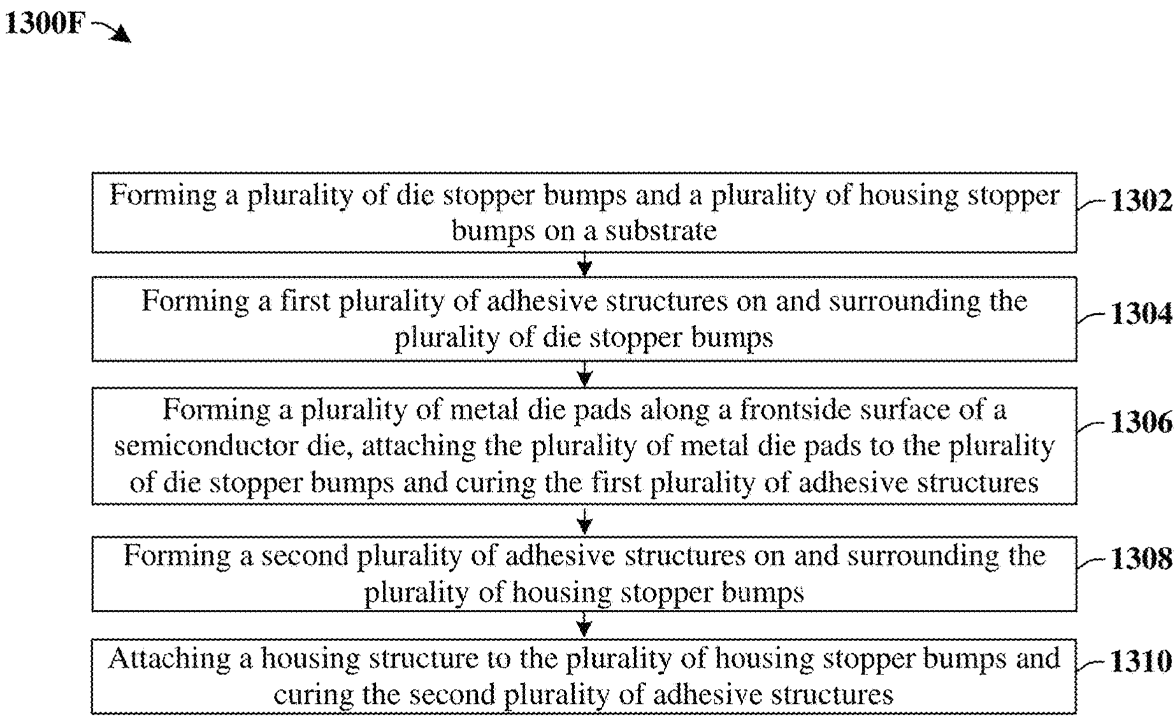
1300F
Forming a plurality of die stopper bumps and a plurality of housing stopper bumps on a substrate
1302
Forming a first plurality of adhesive structures on and surrounding the plurality of die stopper bumps
1304
Forming a plurality of metal die pads along a frontside surface of a semiconductor die, attaching the plurality of metal die pads to the plurality of die stopper bumps and curing the first plurality of adhesive structures
1306
Forming a second plurality of adhesive structures on and surrounding the plurality of housing stopper bumps
1308
Attaching a housing structure to the plurality of housing stopper bumps and curing the second plurality of adhesive structures
1310

Fig. 13F

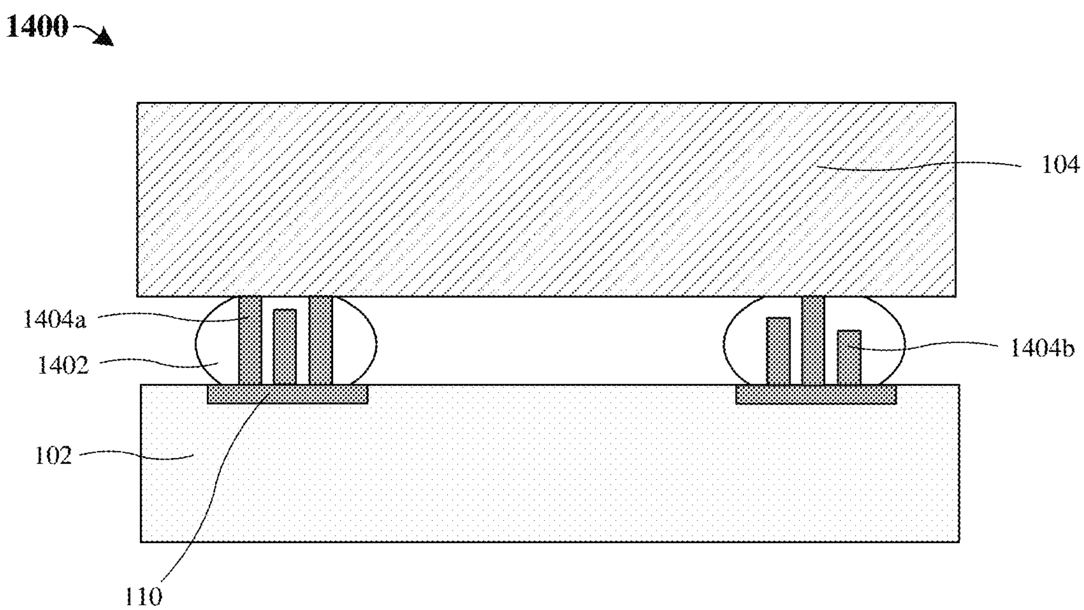

Fig. 14

DIE ATTACHED LEVELING CONTROL BY METAL STOPPER BUMPS

REFERENCE TO RELATED APPLICATION

This Application is a Continuation of U.S. application Ser. No. 17/184,924, filed on Feb. 25, 2021, the contents of which are hereby incorporated by reference in their entirety.

BACKGROUND

Many modern-day integrated chips (ICs) include a semiconductor die that is attached to a substrate or a package, such as a printed circuit board. Thus, printed circuit board can include a mixture of different ICs that are electrically coupled to one another to achieve a higher-level function. For example, a printed circuit board for a mobile phone can include microprocessor ICs to process digital data, imaging ICs to enable camera type functions, analog ICs to regulate power or perform other functions on the mobile phone, and/or micro-electrical-mechanical system (MEMS) ICs to act as an accelerometer and/or global positioning system (GPS) unit, for example.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. **1** illustrates a cross-sectional view **100** of some embodiments of an IC comprising a stopper bump structure.

FIG. **2** illustrates a cross-sectional view of some embodiments of an IC comprising a stopper bump structure separating a housing structure from a semiconductor die.

FIG. **3** illustrates a cross-sectional view of some embodiments of an IC comprising a stopper bump structure separating stacked ICs.

FIG. **4** illustrates a cross-sectional view of some embodiments of an IC comprising a stopper bump structure used to control an angle of a semiconductor die.

FIG. **5** illustrates a cross-sectional view of some embodiments of an IC comprising an electrically conductive stopper bump structure.

FIGS. **6**A-**6**G illustrate a series of cross-sectional and isometric views of some embodiments of a die stopper bump structure.

FIGS. **7**A-**7**F illustrate a series of top views of some embodiments of a stopper bump structure.

FIGS. **8**A-**8**E illustrate a series of top views of some embodiments of a metal pad structure.

FIGS. **9**A-**9**F illustrate a series of cross-sectional views for a method of forming an IC comprising a stopper bump structure.

FIG. **9**G illustrates a flowchart of some embodiments of the method consistent with FIGS. **9**A-**9**F.

FIGS. **10**A-**10**G illustrate a series of cross-sectional views for a method of forming an IC comprising a stopper bump structure separating a housing structure from a semiconductor die.

FIG. **10**H illustrates a flowchart of some embodiments of the method consistent with FIGS. **10**A-**10**G.

FIGS. **11**A-**11**I illustrate a series of cross-sectional views for a method of forming an IC comprising a stopper bump structure separating stacked ICs.

FIG. **11**J illustrates a flowchart of some embodiments of the method consistent with FIGS. **11**A-**11**I.

FIGS. **12**A-**12**F illustrate a series of cross-sectional views for a method of forming an IC comprising a stopper bump structure used to control an angle of a semiconductor die.

FIG. **12**G illustrates a flowchart of some embodiments of the method consistent with FIGS. **12**A-**12**F.

FIGS. **13**A-**13**E illustrate a series of cross-sectional views for a method of forming an IC comprising an electrically conductive stopper bump structure.

FIG. **13**F illustrates a flowchart of some embodiments of the method consistent with FIGS. **13**A-**13**E.

FIG. **14** illustrates a cross-sectional view of some embodiments of an IC comprising a stopper bump structure with differing heights.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

An IC package may adhere a semiconductor die to a substrate by an adhesive structure. Many factors may cause the semiconductor die to tilt at an angle of other than zero degrees with respect to a laterally-extending surface of the substrate. Some of these factors may include a variability in the viscosity of the adhesive structure, an inability to manipulate the semiconductor die after adhesive curing, a variability in the amount of adhesive dispensed, a variability in adhesive shrinkage after curing, a variability in the geometry of the adhesive structure, and/or a variability in contact angle between the adhesive structure and the semiconductor die. Each of these factors may work individually and/or in combination to contribute to die tilt.

For a semiconductor die in an IC package, die tilt may cause a negative impact to device performance. A semiconductor die may comprise an accelerometer, and die tilt may cause the accelerometer to be in an unbalanced position. Thus, because some devices when manufactured may have accelerometers in an unbalanced position, the manufacturing process may employ significant calibration of the devices after packaging, leading to additional time and manufacturing costs. In addition, for optical devices, excessive die tilt can cause the light path for these optical devices to be offset from the optimal angle (e.g., offset from normal), which may lead to worsened device performance. A semiconductor die may be wire bonded to a substrate or a package to improve performance, but die tilt may still lead to additional time and manufacturing costs, as well as a decrease in device performance.

In the present disclosure, an IC package is presented which allows for die attachment control. Stopper bumps in addition to an adhesive structure are arranged between a semiconductor die and a substrate, between a semiconductor die and a package, and/or between a semiconductor die and an additional semiconductor die to constrain the gap between the semiconductor die and the substrate. The addition of stopper bumps allows for stacked devices to have better gap control and better leveling. For example, an optical device may need to have a lens aligned with an optical axis, and hence the addition of stopper bumps allows for a more level optical device with a lens properly aligned with the optical axis. In an additional example, motion sensors may need to control zero-gravitation offset, which is caused by an accelerometer being misaligned. The addition of stopper bumps allows for the accelerometer in the motion sensor to be properly aligned after the packaging process, and additional time and materials are not required to realign the semiconductor die, thus controlling zero-gravitation offset.

FIG. **1** illustrates a cross-sectional view **100** of some embodiments of an IC comprising a stopper bump structure. The IC comprises a substrate **102** underlying a semiconductor die **104**. The substrate **102** may mechanically support and electrically connect electrical components such as the semiconductor die **104** and other electrical components that are not shown. In some embodiments, a frontside surface **104***f* of the semiconductor die **104** is above a backside surface **104***b* of the semiconductor die **104**. The substrate **102** may be or otherwise comprise, for example, an IC die, a printed circuit board (PCB), or some other suitable type of substrate. The semiconductor die **104** may be or otherwise comprise, for example, a MEMS structure, a semiconductor substrate, or some other suitable integrated chip. In some embodiments, the front side surface **104***f* of the semiconductor die **104** generally corresponds to a side of the die on which active devices are disposed, while the back side surface **104***b* corresponds to a side of the die without active devices thereon, however in other embodiments this naming convention can be flipped. The front side surface **104***f* and/or back side surface **104***b* can also be referred to as a first side and second side (respectively, or vice versa).

A first plurality of metal pads **110** are disposed along the top surface of the substrate **102**. A plurality of die stopper bumps **106** separate the substrate **102** from the semiconductor die **104**. In some embodiments, the plurality of die stopper bumps **106** directly contacts the semiconductor die **104** and the first plurality of metal pads **110**. The plurality of die stopper bumps **106** are configured to control an angle of operation of the semiconductor die **104**. A first plurality of adhesive structures **108** may surround each of the plurality of die stopper bumps **106** and provides adhesion between the semiconductor die **104** and the substrate **102**. A second plurality of metal pads **116** are disposed along a frontside surface **104***f* of the semiconductor die **104**.

In some embodiments, a housing structure **114** overlies and provides protection to the semiconductor die **104**. The housing structure **114** may surround outer sidewalls of the semiconductor die **104**. In some embodiments, both an inner surface of the housing structure **114** and an outer surface of the housing structure **114** have a U-shaped cross-sectional profile. The inner surface and the outer surface of the housing structure **114** may be connected by a bottom surface of the housing structure **114**. The bottom surface may be vertically between the top surface of the substrate **102** and a lateral portion of the inner surface of the housing structure **114**.

A third plurality of metal pads **118** and a fourth plurality of metal pads **124** are disposed along a top surface of the substrate **102**. In some embodiments, the fourth plurality of metal pads **124** are laterally between the first plurality of metal pads **110** and the third plurality of metal pads **118**. A plurality of housing stopper bumps **120** separate the substrate **102** from the housing structure **114**. In some embodiments, the plurality of housing stopper bumps **120** directly contacts the housing structure **114** and the third plurality of metal pads **118**. The plurality of housing stopper bumps **120** are configured to control an angle of operation of the housing structure **114**. A second plurality of adhesive structures **122** may surround each of the plurality of housing stopper bumps **120** and provides adhesion between the housing structure **114** and the substrate **102**. In some embodiments, the first plurality of metal pads **110**, the second plurality of metal pads **116**, the third plurality of metal pads **118**, and/or the fourth plurality of metal pads **124** are conductive and serve as a live electrical connection. In some embodiments, the first plurality of metal pads **110**, the second plurality of metal pads **116**, the third plurality of metal pads **118**, and/or the fourth plurality of metal pads **124** are floating without an electrical connection. A metal wire **112** bonds the fourth plurality of metal pads **124** to the second plurality of metal pads **116**. In some embodiments, the metal wire **112** may provide stability to the semiconductor die **104**. In some embodiments, the metal wire **112** may provide an electrical connection between the semiconductor die **104** and the substrate **102**.

The plurality of die stopper bumps **106** and the plurality of housing stopper bumps **120** may be or otherwise comprise, for example, gold, copper, iron, nickel, or some other suitable material(s). The first plurality of metal pads **110**, the second plurality of metal pads **116**, and the third plurality of metal pads **118** may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s). The metal wires **112** may be or otherwise comprise, for example, aluminum, copper, silver, gold, or some other suitable metal(s). The housing structure **114** may be or otherwise, for example, comprise molded plastic, ceramic, or some other suitable packaging material(s). The first plurality of adhesive structures **108** and the second plurality of adhesive structures **122** may be or otherwise comprise, for example, epoxy glue or some other suitable adhesive(s). The plurality of die stopper bumps **106** have first predetermined heights that set the tilt of the semiconductor die **104**, and the plurality of housing stopper bumps **120** have second predetermined heights that set the tilt of the housing structure **114**. In some embodiments, each of the plurality of die stopper bumps **106** share a first height h**1**, wherein h**1** can range from 15 um to 400 um, and each of the plurality of housing stopper bumps **120** share a second height h**2**, wherein h**2** can range from 15 um to 400 um. In some embodiments, h**1** is equal to h**2**. In some embodiments, h**1** is less than or greater than h**2**.

Bonding a semiconductor die to a substrate may cause the semiconductor die to tilt, leading to an impact on device performance and increase in manufacturing cost and time. The IC illustrated in FIG. **1** allows for tilt control of both the semiconductor die **104** and the housing structure **114**, while still adhering the semiconductor die **104** and the housing structure **114** to the substrate **102**. The plurality of die stopper bumps **106** constrain a gap between the semiconductor die **104** and the substrate **102**, allowing for the semiconductor die **104** to have a controlled level of tilt while still adhering to the substrate **102**. The plurality of housing stopper bumps **120** constrain a gap between the housing structure **114** and the substrate **102**, allowing for the housing structure **114** to have a controlled level of tilt while still allowing for the housing structure **114** to be adhered to the substrate **102**. An optical device may utilize precise package leveling control to ensure a proper optical path to the optical device. Controlled tilt of the semiconductor die **104** and the housing structure **114** may have a positive impact on device performance while decreasing manufacturing cost and manufacturing time.

FIG. **2** illustrates a cross-sectional view **200** of some embodiments of an IC comprising a stopper bump structure separating a housing structure from a semiconductor die. The IC comprises a substrate **102** disposed below a semiconductor die **104**. A housing structure **114** is disposed over the substrate **102**. A third plurality of metal pads **118** is disposed along a top surface of the substrate **102**. In some embodiments, a bottom surface of the third plurality of metal pads **118** is below a top surface of the substrate **102**. A bottom surface of the housing structure **114** directly overlies the third plurality of metal pads **118**. A plurality of housing stopper bumps **120** vertically separates the housing structure **114** from the third plurality of metal pads **118**. A second plurality of adhesive structures **122** surrounds each of the plurality of housing stopper bumps **120**, such that each of the second plurality of adhesive structures **122** is a single continuous body. In some embodiments, a frontside surface **104*f*** of the semiconductor die **104** is below a backside surface **104*b*** of the semiconductor die **104**.

A plurality of metal housing pads **202** is disposed along the lateral portion of the inner surface of the housing structure **114**. A plurality of die stopper bumps **204** separates the housing structure **114** from the semiconductor die **104**. The plurality of die stopper bumps **204** vertically separates the semiconductor die **104** from a bottom surface of the plurality of metal housing pads **202**. The plurality of die stopper bumps **204** are configured to control an angle of operation of the semiconductor die **104**. A plurality of housing adhesive structures **206** surround each of the plurality of die stopper bumps **204** and provide adhesion between the semiconductor die **104** and the plurality of metal housing pads **202**. In some embodiments, metal wires **208** connect a bottom surface of the plurality of metal housing pads **202** to a bottom surface of the second plurality of metal pads **116**.

The metal housing pads **202** may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s). The plurality of die stopper bumps **204** may be or otherwise comprise, for example, gold, copper, iron, nickel, or some other suitable material(s). The plurality of housing adhesive structures **206** may be or otherwise comprise, for example, epoxy glue or some other suitable adhesive(s). The metal wires **208** may be or otherwise comprise, for example, aluminum, copper, silver, gold, or some other suitable metal(s).

The IC illustrated in FIG. **2** allows for tilt control of both the semiconductor die **104** and the housing structure **114**, while still adhering the semiconductor die **104** to the housing structure **114** and the housing structure **114** to the substrate **102**. The plurality of die stopper bumps **204** constrain a gap between the semiconductor die **104** and the housing structure **114**, allowing for the semiconductor die **104** to have a controlled level of tilt while still adhering to the housing structure **114**. Controlled tilt of the semiconductor die **104** through the use of the plurality of die stopper bumps **204** may have a positive impact on device performance while decreasing manufacturing cost and manufacturing time.

FIG. **3** illustrates a cross-sectional view **300** of some embodiments of an IC comprising a stopper bump structure separating stacked ICs. A semiconductor die **104** is disposed above a substrate **102**. A first plurality of metal pads **110** are disposed between the substrate **102** and the semiconductor die **104**. In some embodiments, a frontside surface **104*f*** of the semiconductor die **104** is above a backside surface **104*b*** of the semiconductor die **104**.

A plurality of die stopper bumps **106** separates the substrate **102** from the semiconductor die **104**. In some embodiments, a top surface of the plurality of die stopper bumps **106** contacts the backside surface **104*b*** of the semiconductor die **104**, and a bottom surface of the plurality of die stopper bumps **106** contacts a top surface of the first plurality of metal pads **110**. A first plurality of adhesive structures **108** is disposed between the first plurality of metal pads **110** and the semiconductor die **104**. In some embodiments, each of the plurality of die stopper bumps **106** is laterally surrounded by an adhesive structure of the plurality of adhesive structures **108**.

A plurality of metal die pads **310** overlies the semiconductor die **104**. A stacked semiconductor die **304** overlies the plurality of metal die pads **310**. In some embodiments, outer sidewalls of the stacked semiconductor die **304** may be laterally offset from outer sidewalls of the semiconductor die **104**. A plurality of overlying die stopper bumps **308** separate the stacked semiconductor die **304** from the semiconductor die **104**. In some embodiments, the plurality of overlying die stopper bumps **308** directly contact the stacked semiconductor die **304** and the plurality of metal die pads **310**. The plurality of overlying die stopper bumps **308** is configured to control an angle of operation of the stacked semiconductor die **304**.

A plurality of overlying adhesive structures **306** may surround each of the plurality of overlying die stopper bumps **308**. In some embodiments, each of the plurality of overlying die stopper bumps **308** is laterally surrounded by an adhesive structure of the plurality of overlying adhesive structures **306**. In some embodiments, one or more overlying adhesive structure of the plurality of overlying adhesive structures **306** may continuously extend to surround more than one die stopper bump of the plurality of overlying die stopper bumps **308**. In some embodiments, each of the plurality of overlying adhesive structures **306** continuously extends to surround no more than one overlying die stopper bump of the plurality of overlying die stopper bumps **308**.

In some embodiments, a housing structure **302** overlies and provides protection to the semiconductor die **104** and the stacked semiconductor die **304**. The housing structure **302** may surround outer sidewalls of the semiconductor die **104** and outer sidewalls of the stacked semiconductor die **304**. In some embodiments, both an inner surface of the housing structure **302** and an outer surface of the housing structure **302** have a U-shaped cross-sectional profile. The inner surface and the outer surface of the housing structure **302** may be connected by a bottom surface of the housing structure **302**. The bottom surface may be vertically between the top surface of the substrate **102** and a lateral portion of the inner surface of the housing structure **302**.

A third plurality of metal pads **118** and a fourth plurality of metal pads **124** are disposed along the top surface of the substrate **102**. A plurality of housing stopper bumps **120** separates the substrate **102** from the bottom surface of a housing structure **302**. In some embodiments, the plurality of housing stopper bumps **120** directly contacts the housing structure **302** and the third plurality of metal pads **118**. The plurality of housing stopper bumps **120** are configured to control an angle of operation of the housing structure **302**. A second plurality of adhesive structures **122** may surround each of the plurality of housing stopper bumps **120** and provides adhesion between the housing structure **302** and the substrate **102**. A metal wire **112** couples the fourth plurality of metal pads **124** to the second plurality of metal pads **116**.

The stacked semiconductor die **304** may be or otherwise comprise, for example, a MEMS structure, a semiconductor substrate, or some other suitable integrated chip. In some embodiments, the stacked semiconductor die **304** and the semiconductor die **104** may be structurally similar. In some embodiments, the stacked semiconductor die **304** and the semiconductor die **104** may be structurally different. The plurality of overlying die stopper bumps **308** may be or otherwise comprise, for example, gold, copper, iron, nickel, or some other suitable material(s). The plurality of metal die pads **310** may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s). The housing structure **302** may be or otherwise, for example, comprise molded plastic, ceramic, or some other suitable packaging material(s). The plurality of overlying adhesive structures **306** may be or otherwise comprise, for example, epoxy glue or some other suitable adhesive(s).

The IC illustrated in FIG. **3** allows for tilt control of the semiconductor die **104**, the housing structure **302** and the stacked semiconductor die **304** while still adhering the semiconductor die **104** and the housing structure **302** to the substrate **102**, and a bottom surface of the stacked semiconductor die **304** to the frontside surface **104*f*** of the semiconductor die **104**. The plurality of die stopper bumps **106** constrain a gap between the semiconductor die **104** and the substrate **102**. Further, the plurality of overlying die stopper bumps **308** constrain a gap between the semiconductor die **104** and the stacked semiconductor die **304**. This decreases manufacturing cost and manufacturing time, and allows for three-dimensional integrated chips (3D-ICs) to have tilt control between semiconductor dies.

FIG. **4** illustrates a cross-sectional view **400** of some embodiments of an IC comprising a stopper bump structure used to control an angle of a semiconductor die. The IC comprises a substrate **102** underlying a semiconductor die **104**. In some embodiments, a frontside surface of the semiconductor die **104** is above a backside surface **104*b*** of the semiconductor die **104**.

A first plurality of metal pads **110** are disposed along the top surface of the substrate **102**, such that sidewalls of each of the plurality of metal pads **110** contacts inner sidewalls of the substrate **102**. A first plurality of die stopper bumps **406** are disposed on a first end of the backside of the semiconductor die **104** and separate the substrate **102** from the semiconductor die **104**. In some embodiments, the first plurality of die stopper bumps **406** directly contacts the semiconductor die **104** and the first plurality of metal pads **110**. The first plurality of die stopper bumps **406** may have a height h**3**. A first plurality of die adhesive structures **408** may surround each of the first plurality of die stopper bumps **406** and provides adhesion between the semiconductor die **104** and the substrate **102**.

A second plurality of die stopper bumps **402** are disposed on a second end of the backside **104*b*** of the semiconductor die **104** opposite the first end, such that the first plurality of die stopper bumps **406** and the second plurality of die stopper bumps **402** are configured to keep a backside surface **104*b*** of the semiconductor die **104** above a top surface of the first plurality of metal pads **110**. The second plurality of die stopper bumps **402** separates the substrate **102** from the semiconductor die **104**. In some embodiments, the second plurality of die stopper bumps **402** directly contacts the semiconductor die **104** and the first plurality of metal pads **110**. The second plurality of die stopper bumps **402** may have a height h**4**. In some embodiments, h**4** is larger than h**3**. A second plurality of die adhesive structures **404** may surround each of the second plurality of die stopper bumps **402** and provides adhesion between the semiconductor die **104** and the substrate **102**. In some embodiments, the first plurality of die stopper bumps **406** and the second plurality of die stopper bumps **402** are configured to control a tilt angle A**1** of the semiconductor die **104**. In some embodiments, the tilt angle A**1** of the semiconductor die **104** is greater than 0 degrees with respect to a top surface of the substrate **102**.

A second plurality of metal pads **116** are disposed along a frontside surface **104*f*** of the semiconductor die **104**. In some embodiments, the second plurality of metal pads **116** directly overlie the first plurality of metal pads **110**. In some embodiments, a housing structure **114** is disposed over the semiconductor die **104**.

A third plurality of metal pads **118** and a fourth plurality of metal pads **124** are disposed along a top surface of the substrate **102**. A plurality of housing stopper bumps **120** are disposed vertically between a bottom surface of the housing structure **114** and a top surface of the third plurality of metal pads. In some embodiments, each of the plurality of housing stopper bumps **120** may be at a different height to control an angle of the housing structure **114** to range from approximately 0 degrees to approximately 20 degrees with respect to a top surface of the substrate **102**. A second plurality of adhesive structures **122** may surround each of the plurality of housing stopper bumps **120**. A metal wire **112** mechanically connects the fourth plurality of metal pads **124** to the second plurality of metal pads **116**. In some embodiments, the metal wire **112** may provide stability to the semiconductor die **104**. In some embodiments, the metal wire **112** may provide an electrical connection between the semiconductor die **104** and the substrate **102**.

The first plurality of die stopper bumps **402** and the second plurality of die stopper bumps **406** may be or otherwise comprise, for example, gold, copper, iron, nickel, or some other suitable material(s). The first plurality of die adhesive structures **408** and the second plurality of die adhesive structures **404** may be or otherwise comprise, for example, epoxy glue or some other suitable adhesive(s).

The IC illustrated in FIG. **4** allows for tilt control of the semiconductor die **104** to a desired angle of operation A**1**, while adhering the semiconductor die **104** to the substrate **102**. The first plurality of die stopper bumps **406** and the second plurality of die stopper bumps **402** are at different heights h**3** and h**4**, allowing for the semiconductor die **104** to have a controlled angle of tilt A**1** while still adhering to the substrate **102**. Controlled tilt of the semiconductor die

104 to a specific tilt angle A1 may have a positive impact on device performance in certain devices that may benefit from a tilted angle of operation.

FIG. 5 illustrates a cross-sectional view 500 of some embodiments of an IC comprising an electrically conductive stopper bump structure. The IC comprises a substrate 102 underlying a semiconductor die 104. The substrate 102 may mechanically support and electrically connect electrical components such as the semiconductor die 104 and other electrical components that are not shown. In some embodiments, a frontside surface of the semiconductor die 104 is below a backside surface of the semiconductor die 104.

A plurality of die stopper bumps 106 vertically separate the substrate 102 from the semiconductor die 104. In some embodiments, the plurality of die stopper bumps 106 separates the semiconductor die 104 and a first plurality of metal pads 110 overlying the substrate 102. A first plurality of adhesive structures 108 may laterally surround each of the plurality of die stopper bumps 106. A plurality of metal die pads 502 is disposed along a frontside surface of the semiconductor die 104 and is configured to electrically couple the semiconductor die 104 to the substrate 102. In some embodiments, the plurality of die stopper bumps 106 are electrically conductive and act as vias electrically coupling the first plurality of metal pads 110 to the plurality of metal die pads 502. The plurality of metal die pads 502 may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s).

In some embodiments, a housing structure 114 is disposed directly over the substrate 102. The housing structure 114 may surround the semiconductor die 104. A plurality of housing stopper bumps 120 separates the housing structure 114 from a third plurality of metal pads 118 overlying the substrate 102. A second plurality of adhesive structures 122 may laterally surround each of the plurality of housing stopper bumps 120.

The IC illustrated in FIG. 5 allows for tilt control of both the semiconductor die 104 and the housing structure 114, while further allowing for an electrical current to pass between the substrate 102 and the semiconductor die 104. The plurality of die stopper bumps 106 constrain a gap between the semiconductor die 104 and the substrate 102, allowing for the semiconductor die 104 to have a controlled level of tilt while still adhering to the substrate 102. Further, the plurality of metal die pads 502 may be electrically conductive, allowing for the semiconductor die 104 to be electrically coupled to the substrate 102 by the plurality of die stopper bumps 106. Controlled tilt of the semiconductor die 104 by means of electrically conductive vias may decrease manufacturing cost and manufacturing time, while allowing for further applications.

With reference to FIGS. 6A-6G, a series of cross-sectional and isometric views 600A-600G illustrate some embodiments of a die stopper bump structure. The die stopper bump structures may, for example, correspond to the die stopper bumps of FIG. 1.

FIG. 6A illustrates a cross-sectional view 600A of some embodiments of a die stopper bump structure. The die stopper bump structure comprises a plurality of stud bumps 602, each comprising an upper portion 602*u* and a lower portion 6021. The plurality of stud bumps 602 vertically separate a substrate 102 from a semiconductor die 104. A metal pad 110 overlies the substrate 102, and the plurality of stud bumps 602 are disposed on the metal pad 110. An adhesive structure 108 surrounds outer sidewalls of each of the plurality of stud bumps 602. In some embodiments, a bottom surface of the lower portion 6021 of each stud bump 602 directly contacts the metal pad 110. In some embodiments, a top surface of the upper portion 602*u* of each stud bump 602 directly contacts the semiconductor die 104. In some embodiments, the plurality of stud bumps 602 may serve as an electrical interconnect between the semiconductor die 104 and the substrate 102. In some embodiments, the plurality of stud bumps 602 may separate the substrate 102 from a housing structure. In some embodiments, the plurality of stud bumps 602 may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s).

FIG. 6B illustrates an isometric view 600B of some embodiments of a stud bump. The stud bump may, for example, correspond to one of the plurality of stud bumps illustrated in FIG. 6A. The stud bump 602 comprises an upper portion 602*u* and a lower portion 6021. A height h5 of each stud bump 602 may range from 35 um to 60 um. In some embodiments, a height of the lower portion 6021 of each stud bump 602 may equal a height of the upper portion 602*u*. The upper portion 602*u* of each stud bump 602 may have a diameter d1 ranging from 15 um to 35 um. The lower portion 6021 of each stud bump 602 may have a diameter d2 ranging from 50 um to 90 um.

FIG. 6C illustrates a cross-sectional view 600C of some embodiments of a die stopper bump structure. The die stopper bump structure comprises a metal ball bump 608 that separates a substrate 102 from a semiconductor die 104. A metal pad 110 overlies the substrate, and the metal ball bump 608 overlies the metal pad 110. In some embodiments, the metal pad 110 may directly contact a bottom surface of the metal pad 110. A metal ball adhesive structure 108 surrounds outer sidewalls of the metal ball bump 608. In some embodiments, a height of the metal ball bump 608 may range from 250 um to 400 um. In some embodiments, the die stopper bump structure may comprise a plurality of metal ball bumps to achieve better stabilization of the semiconductor die 104. In some embodiments, the plurality of metal ball bumps 608 may separate the substrate 102 from a housing structure. In some embodiments, the plurality of metal ball bumps 608 may be or otherwise comprise, for example, tin, lead, copper, iron, nickel, gold, silver, some other suitable metal(s), or a combination of the foregoing.

FIG. 6D illustrates a cross-sectional view 600D of some embodiments of a die stopper bump structure. The die stopper bump structure comprises a plurality of stud bumps 610, each comprising an upper portion 610*u* and a plurality of stacked lower portions 6101. The plurality of stud bumps 610 vertically separate a substrate 102 from a semiconductor die 104. A metal pad 110 overlies the substrate 102, and the plurality of stud bumps 610 are disposed on the metal pad 110. An adhesive structure 108 surrounds outer sidewalls of each of the plurality of stud bumps 610. In some embodiments, a bottom surface of the plurality of stacked lower portions 6101 of each stud bump 610 directly contacts the metal pad 110. In some embodiments, a top surface of the upper portion 610*u* of each stud bump 610 directly contacts the semiconductor die 104. In some embodiments, the plurality of stacked lower portions 6101 comprise at least three stacked portions. In some embodiments, the plurality of stacked lower portions 6101 have a maximum width greater than that of the upper portion 610*u*. In some embodiments, the plurality of stud bumps 610 may have the height h6. In some embodiments, the height h6 may range from 65 um to 120 um. In some embodiments, the plurality of stacked lower portions 6101 may have the diameter d2, and the upper portion 610*u* may have the diameter d1. In some embodiments, the plurality of stud bumps 610 may separate the substrate **102** from a housing structure. In some embodiments, the plurality of stud bumps **610** may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s).

FIG. **6**E illustrates a cross-sectional view **600**E of some embodiments of a die stopper bump structure. The die stopper bump structure comprises a plurality of stud bumps **612**. The plurality of stud bumps **612** vertically separate a substrate **102** from a semiconductor die **104**. A metal pad **110** overlies the substrate **102**, and the plurality of stud bumps **612** are disposed on the metal pad **110**. An adhesive structure **108** surrounds outer sidewalls of each of the plurality of stud bumps **612**. In some embodiments, a bottom surface of the plurality of stud bumps **612** directly contacts the metal pad **110**. In some embodiments, a top surface of the plurality of stud bumps **612** directly contacts the semiconductor die **104**. In some embodiments, the plurality of stud bumps **612** have a height ranging from 15 um to 25 um, and a diameter ranging from 75 um to 85 um. In some embodiments, the plurality of stud bumps **612** have a uniform height extending from the bottom surface of the plurality of stud bumps **612** to the top surface of the plurality of stud bumps **612**. In some embodiments, the plurality of stud bumps **612** may separate the substrate **102** from a housing structure. In some embodiments, the plurality of stud bumps **612** may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s).

FIG. **6**F illustrates a cross-sectional view **600**F of some embodiments of a die stopper bump structure. The die stopper bump structure comprises a plurality of stud bumps **614** with a curved top surface. The plurality of stud bumps **614** vertically separate a substrate **102** from a semiconductor die **104**. A metal pad **110** overlies the substrate **102**, and the plurality of stud bumps **614** are disposed on the metal pad **110**. An adhesive structure **108** surrounds outer sidewalls of each of the plurality of stud bumps **614**. In some embodiments, a bottom surface of the plurality of stud bumps **614** directly contacts the metal pad **110**. In some embodiments, a top surface of the plurality of stud bumps **614** directly contacts the semiconductor die **104**. In some embodiments, the plurality of stud bumps **614** may serve as an electrical interconnect between the semiconductor die **104** and the substrate **102**. In some embodiments, the plurality of stud bumps **614** have a height ranging from 15 um to 25 um, and a diameter ranging from 90 um to 110 um. In some embodiments, the plurality of stud bumps **614** may separate the substrate **102** from a housing structure. In some embodiments, the plurality of stud bumps **614** may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s).

FIG. **6**G illustrates a cross-sectional view **600**G of some embodiments of a die stopper bump structure. The die stopper bump structure comprises a plurality of stud bumps **614**, each comprising an upper portion **614***u* and a lower portion **6141**. The upper portion **614***u* has a cross-sectional profile that is a truncated cone, wherein a bottom surface of the truncated cone faces the lower portion **6141** and wherein a top surface of the truncated cone directly contacts the semiconductor die **104**. The plurality of stud bumps **614** vertically separate a substrate **102** from a semiconductor die **104**. A metal pad **110** overlies the substrate **102**, and the plurality of stud bumps **614** are disposed on the metal pad **110**. An adhesive structure **108** surrounds outer sidewalls of each of the plurality of stud bumps **614**. In some embodiments, a bottom surface of the lower portion **6141** of each stud bump **614** directly contacts the metal pad **110**. In some embodiments, the plurality of stud bumps **614** have a height greater than the height h**5**. In some embodiments, the lower portion **6141** may have the diameter d**2**, and the upper portion **614***u* may have the diameter d**1**. In some embodiments, the plurality of stud bumps **614** may separate the substrate **102** from a housing structure. In some embodiments, the plurality of stud bumps **614** may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s).

With reference to FIGS. **7**A-**7**F, a series of top views **700**A-**700**F illustrate some embodiments of a stopper bump structure. The stopper bump structures may, for example, correspond to the die stopper bumps of FIG. **1**.

FIG. **7**A illustrates a top view **700**A of some embodiments of a stopper bump structure. A metal pad **110** is disposed over a substrate **102**. A stopper bump structure **702** is disposed over the metal pad **110**, and an adhesive structure **108** surrounds the stopper bump structure **702**. The stopper bump structure **702** comprises a single stopper bump disposed at the center of the metal pad **110**. In some embodiments, the adhesive structure **108** uniformly surrounds the stopper bump structure **702**. The metal pad **110** has a length L**1** and a width w**1**. In some embodiments, the stopper bump structure **702** is a stud bump and the length L**1** and the width w**1** of the metal pad **110** both range from 100 um to 150 um. In some embodiments, the stopper bump structure is a metal ball bump and the length L**1** and the width w**1** of the metal pad **110** both range from 300 um to 450 um. The stopper bump structure **702** has a diameter d**1**. In some embodiments, the stopper bump structure is a stud bump and the diameter d**1** ranges from 50 um to 100 um. In some embodiments, the stopper bump structure is a metal ball bump and the diameter d**1** ranges from 250 um to 400 um. In some embodiments, the stopper bump structure **702** is more than 20 um away from any edge of the metal pad **110**. In some embodiments, the stopper bump structure **702** may be or otherwise comprise, for example, tin, lead, copper, iron, nickel, gold, silver, some other suitable metal(s), or a combination of the foregoing.

FIG. **7**B illustrates a top view **700**B of some embodiments of a stopper bump structure. A metal pad **110** is disposed over a substrate **102**, a stopper bump structure **702** is disposed over the metal pad **110**, and an adhesive structure **108** surrounds the stopper bump structure **702**. The stopper bump structure **702** comprises a plurality stopper bumps, such that each stopper bump is separated in a diagonal direction. The metal pad **110** has a length L**1** and a width w**1**. In some embodiments, the stopper bump structure **702** is a stud bump and the length L**1** and the width w**1** of the metal pad **110** both range from 150 um to 250 um. In some embodiments, the stopper bump structure is a metal ball bump and the length L**1** and the width w**1** of the metal pad **110** both range from 550 um to 950 um. The stopper bump structure **702** has a diameter d**4**. In some embodiments, the stopper bump structure is a stud bump and the diameter d**4** ranges from 50 um to 100 um. In some embodiments, the stopper bump structure is a metal ball bump and the diameter d**4** ranges from 250 um to 400 um. In some embodiments, the individual stopper bumps of the stopper bump structure **702** are separated by a distance d**5** that is greater than 20 um. In some embodiments, the stopper bump structure **702** is greater than 20 um away from any edge of the metal pad **110**.

FIG. **7**C illustrates a top view **700**C of some embodiments of a stopper bump structure. A stopper bump structure **702** is disposed over a metal pad **110**, which is disposed over a substrate **102**. An adhesive structure **108** surrounds the stopper bump structure **702**. The stopper bump structure **702** comprises a plurality of rows and a plurality of columns of stopper bumps. The metal pad **110** has a length L**1** and a width w**1**. In some embodiments, the stopper bump structure **702** is a stud bump and the length L**1** and the width w**1** of the metal pad **110** both range from 150 um to 250 um. In some embodiments, the stopper bump structure is a metal ball bump and the length L**1** and the width w**1** of the metal pad **110** both range from 550 um to 950 um. The stopper bump structure **702** has a diameter d**4**. In some embodiments, the stopper bump structure is a stud bump and the diameter d**4** ranges from 50 um to 100 um. In some embodiments, the stopper bump structure is a metal ball bump and the diameter d**4** ranges from 250 um to 400 um. In some embodiments, the individual stopper bumps of the stopper bump structure **702** are separated by a diagonal distance d**5** that is no less than 20 um, and a lateral distance d**6** that is no less than 20 um. In some embodiments, the stopper bump structure **702** is no less than 20 um away from any edge of the metal pad **110**.

FIG. **7**D illustrates a top view **700**D of some embodiments of an elongated stopper bump structure. A stopper bump structure **708** is disposed over an elongated metal pad **706** that has a greater length than width. An adhesive structure **108** surrounds the stopper bump structure **702**. The stopper bump structure **702** comprises a plurality of stopper bumps aligned along a width-wise center of the elongated metal pad **706** in a single row. In some embodiments, the stopper bump structure **708** is approximately 20 um to 30 um away from any edge of the metal pad **706**. In some embodiments, the stopper bumps of the stopper bump structure **708** are approximately 20 um to 30 um away from each other. In some embodiments, the stopper bump structure **708** may be or otherwise comprise, for example, tin, lead, copper, iron, nickel, gold, silver, some other suitable metal(s), or a combination of the foregoing. The metal pad **706** may be or otherwise comprise, for example, copper, iron, nickel, gold, silver, or some other suitable metal(s).

FIG. **7**E illustrates a top view **700**E of some embodiments of an elongated stopper bump structure. A stopper bump structure **708** is disposed over an elongated metal pad **706** that has a greater length than width. An adhesive structure **108** surrounds the stopper bump structure **702**. The stopper bump structure **702** comprises a plurality of stopper bumps aligned in a plurality of rows and a plurality of columns of stopper bumps, such that each row comprises a single stopper bump, and the individual stopper bumps of the stopper bump structure **708** are separated diagonally from each other.

FIG. **7**F illustrates a top view **700**F of some embodiments of an elongated stopper bump structure. A stopper bump structure **708** is disposed over an elongated metal pad **706** that has a greater length than width. An adhesive structure **108** surrounds the stopper bump structure **702**. The stopper bump structure **702** comprises a plurality of stopper bumps aligned in a plurality of rows and a plurality of columns of stopper bumps.

With reference to FIGS. **8**A-**8**E, a series of top views **800**A-**800**E illustrate some embodiments of a metal pad structure that may underlie a plurality of die stopper bumps. The metal pad structure may, for example, correspond to the first plurality of metal pads **110** of FIG. **1**. For example, in the previously described example of FIG. **1**, metal pads **110** when viewed from a top view could be consistent with discrete metal pads **802** and elongated metal pads **808** of FIG. **8**E; such that the semiconductor die **104** of FIG. **1** may have an outer boundary **104***ob* that lies outside of an outer boundary defined by discrete metal pads **802** and elongated metal pads **808** as shown in FIGS. **8**A-**8**E.

FIG. **8**A illustrates a top view **800**A of some embodiments of a metal pad structure. A plurality of discrete metal pads **802** overlies a peripheral region of a substrate **102**. In some embodiments, each of the plurality of discrete metal pads **802** is square-shaped. An overlying semiconductor die would have an outer boundary outside of an outer boundary of the plurality of discrete metal pads **802**.

FIG. **8**B illustrates a top view **800**B of some embodiments of a metal pad structure. A continuous metal pad **804** overlies a peripheral region of a substrate **102**. In some embodiments, the continuous metal pad **804** is ring-shaped with an inner surface and an outer surface that are both square shaped. An overlying semiconductor die would have an outer boundary outside of the outer surface of the continuous metal pad **804**.

FIG. **8**C illustrates a top view **800**C of some embodiments of a metal pad structure. A plurality of elongated metal pads **808** and a plurality of discrete metal pads **802** overlie a peripheral region of a substrate **102**. In some embodiments, each of the elongated metal pads **808** are rectangular. In some embodiments, each of the plurality of discrete metal pads **802** are square-shaped. An overlying semiconductor die would have an outer boundary outside of an outer boundary of the plurality of elongated metal pads **808** and an outer boundary of the plurality of discrete metal pads **802**. In some embodiments, the elongated metal pads **808** alternate with the discrete metal pads **802** around a periphery of the substrate **102** in a square-shaped pattern.

FIG. **8**D illustrates a top view **800**D of some embodiments of a metal pad structure. A plurality of central elongated metal pads **810** and a plurality of central discrete metal pads **812** overlie a central region of a substrate **102**. The central elongated metal pads **810** are elongated parallel to each other. In some embodiments, the plurality of central discrete metal pads **812** are laterally between the plurality of central elongated metal pads **810**, such that inner sidewalls of each of the plurality of central elongated metal pads **810** is facing the plurality of central discrete metal pads **812**. In some embodiments, each of the central elongated metal pads **810** are rectangular. In some embodiments, each of the central discrete metal pads **812** are square. In some embodiments, the plurality of central discrete metal pads **812** are arranged into a plurality of columns and a plurality of rows. In some embodiments, the number of columns is equal to the number of rows.

FIG. **8**E illustrates a top view **800**E of some embodiments of a metal pad structure. A plurality of central elongated metal pads **810** and a plurality of central discrete metal pads **812** overlie a central region of a substrate **102**. A plurality of elongated metal pads **808** and a plurality of discrete metal pads **802** overlie a peripheral region of a substrate **102**. The central elongated metal pads **810** are elongated parallel to each other. The elongated metal pads **808** are elongated parallel to each other. In some embodiments, outer sidewalls of the central elongated metal pads **810** face the plurality of discrete metal pads **810**. In some embodiments, the elongated metal pads **808** alternate with the plurality of discrete metal pads **802** on two opposing ends of the substrate **102**. In some embodiments, each of the central elongated metal pads **810** and each of the elongated metal pads **808** are rectangular. In some embodiments, a length of the central elongated metal pads **810** is greater than that of the elongated metal pads **808**. In some embodiments, each of the central discrete metal pads **812** and each of the discrete metal pads **802** are square. An overlying semiconductor die would have an outer boundary outside of an outer boundary of the plurality of elongated metal pads **808** and an outer boundary of the plurality of discrete metal pads **802**.

With reference to FIGS. **9**A-**9**F, a series of cross sections **900**A-**900**F illustrate some embodiments of a method for forming an IC comprising a stopper bump structure. The IC may, for example, correspond to the IC of FIG. **1**. Although FIGS. **9**A-**9**F are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **9**A-**9**F are not limited to such a method, but instead may stand alone as structures independent of the method.

As illustrated by the cross-sectional view **900**A of FIG. **9**A, a substrate **102** is provided. A first plurality of metal pads **110**, a third plurality of metal pads **118**, and a fourth plurality of metal pads **124** are formed on a top surface of the substrate **102** through a bond pad formation process. The bond pad formation process may be or otherwise comprise, for example, photolithography, deposition of a material through chemical-vapor deposition (CVD), physical-vapor deposition (PVD), electroplating, metal printing (3D printing), or some other suitable deposition process, and chemical-mechanical planarization (CMP) of excess material. A plurality of die stopper bumps **106** is formed on a top surface of the first plurality of metal pads **110** at a first height h**1** by a stopper bump formation process. In some embodiments, the stopper bump formation process may comprise using a wire bonder to provide a stopper bump material onto a metal pad to form ball bond or wedge bond, and then trimming the top of the ball bond or wedge bond (and/or wire at the of the ball bond or wedge bond) to form the plurality of die stopper bumps **106**. A plurality of housing stopper bumps **120** is formed on a top surface of the third plurality of metal pads **118** at a second height h**2** by the stopper bump formation process.

As illustrated by the cross-sectional view **900**B of FIG. **9**B, a first plurality of adhesive structures **108** is formed over the first plurality of metal pads **110** by the stopper bump formation process such that the adhesive structures **108** surround each of the plurality of die stopper bumps **106**. For instance, the first plurality of adhesive structures **108** may be formed by pumping, squeezing, or otherwise providing a liquid around the plurality of die stopper bumps **106** and onto the first plurality of metal pads **110**.

As illustrated by the cross-sectional view **900**C of FIG. **9**C, a semiconductor die **104** is provided. The semiconductor die **104** includes a second plurality of metal pads **116** along a frontside surface of the semiconductor die **104**. A backside surface of the semiconductor die **104** is attached to the plurality of die stopper bumps **106** such that the semiconductor die **104** overlies the substrate **102**. After the semiconductor die **104** is attached, the first plurality of adhesive structures **108** is cured so the liquid of the first plurality of adhesive structures **108** hardens to a solid material.

As illustrated by the cross-sectional view **900**D of FIG. **9**D, metal wires **112** are formed, connecting the second plurality of metal pads **116** to the fourth plurality of metal pads **124**. In some embodiments, the metal wires **112** may be formed with a process similar to that used in the formation of the plurality of die stopper bumps **106**, such that a size of the plurality of die stopper bumps **106** may be correlated to a thickness of the metal wires **112**.

As illustrated by the cross-sectional view **900**E of FIG. **9**E, a second plurality of adhesive structures **122** is formed over the third plurality of metal pads **118** such that the adhesive structures **122** surround each of the plurality of housing stopper bumps **120**. For instance, the second plurality of adhesive structures **122** may be formed by pumping, squeezing, or otherwise providing a liquid around the plurality of housing stopper bumps **120** and onto the third plurality of metal pads **118**.

As illustrated by the cross-sectional view **900**F of FIG. **9**F, a housing structure **114** is provided. A bottom surface of the housing structure **114** is then placed into contact with the liquid of the second plurality of adhesive structures **122** and is pressed down until the housing stopper bumps **120** are firmly positioned between the substrate **102** and semiconductor die **104**. Thus, the housing structure **114** is attached to the plurality of housing stopper bumps **120** such that the housing structure **114** overlies and is vertically separated from the substrate **102**. After the housing structure **114** is attached, the second plurality of adhesive structures **122** is cured.

With respect to FIG. **9**G, a flowchart **900**G of some embodiments of a method for forming an IC comprising a stopper bump structure is illustrated. The IC may, for example, correspond to the IC of FIGS. **9**A-**9**F.

While flowchart **900**G is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

At **902**, a plurality of die stopper bumps and a plurality of housing stopper bumps are formed on a substrate. See, for example, FIG. **9**A.

At **904**, a first plurality of adhesive structures is formed on and surrounds the plurality of die stopper bumps. See, for example, FIG. **9**B.

At **906**, a semiconductor die is attached to the plurality of die stopper bumps, and the first plurality of adhesive structures is cured. See, for example, FIG. **9**C.

At **908**, a frontside surface of the semiconductor die is bonded to the substrate with a metal wire. See, for example, FIG. **9**D.

At **910**, a second plurality of adhesive structures is formed on and surrounds the plurality of housing stopper bumps. See, for example, FIG. **9**E.

At **912**, a housing structure is attached to the plurality of housing stopper bumps, and the second plurality of adhesive structures is cured. See, for example, FIG. **9**F.

With reference to FIGS. **10**A-**10**G, a series of cross sections **1000**A-**1000**G illustrate some embodiments of a method for forming an IC comprising a stopper bump structure separating a housing structure from a semiconductor die. The IC may, for example, correspond to the IC of FIG. **2**. Although FIGS. **10**A-**10**G are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **10**A-**10**G are not limited to such a method, but instead may stand alone as structures independent of the method.

As illustrated by the cross-sectional view **1000**A of FIG. **10**A, a housing structure **114** is provided. A plurality of metal housing pads **202** are formed along a lateral inner surface of the housing structure **114** by the bond pad formation process, such that a first surface of the plurality of metal housing pads **202** directly contacts the housing structure **114**. A plurality of die stopper bumps **204** is formed on second surface of the plurality of metal housing pads **202** that is opposite the first surface by the stopper bump formation process.

As illustrated by the cross-sectional view **1000**B of FIG. **10**B, a plurality of housing adhesive structures **206** formed along the second surface of the plurality of metal housing pads **202** such that the housing adhesive structures **206** surround each of the plurality of die stopper bumps **204**. For instance, the first plurality of housing adhesive structures **206** may be formed by pumping, squeezing, or otherwise providing a liquid around the plurality of die stopper bumps **204** and onto the plurality of metal housing pads **202**.

As illustrated by the cross-sectional view **1000**C of FIG. **10**C, a semiconductor die **104** is provided. The semiconductor die **104** includes a second plurality of metal pads **116** along a frontside surface of the semiconductor die **104**. A backside surface of the semiconductor die **104** is then attached to the plurality of die stopper bumps **204** such that the semiconductor die **104** comprises sidewalls surrounded by the housing structure **114**. After the semiconductor die **104** is attached, the plurality of adhesive structures **206** is cured.

As illustrated by the cross-sectional view **1000**D of FIG. **10**D, metal wires **112** are formed, connecting the second plurality of metal pads **116** to the plurality of metal housing pads **202**. In some embodiments, the metal wires **112** may be formed with a process similar to that used in the formation of the plurality of die stopper bumps **204**, such that a size of the plurality of die stopper bumps **204** may be correlated to a thickness of the metal wires **112**.

As illustrated by the cross-sectional view **1000**E of FIG. **10**E, a substrate **102** is provided. A third plurality of metal pads **118** is formed on a top surface of the substrate **102** by the bond pad formation process. A plurality of housing stopper bumps **120** is formed on a top surface of the third plurality of metal pads **118** by the stopper bump formation process.

As illustrated by the cross-sectional view **1000**F of FIG. **10**F, a second plurality of adhesive structures **122** is formed over the third plurality of metal pads **118** such that the adhesive structures **122** surround each of the plurality of housing stopper bumps **120**.

As illustrated by the cross-sectional view **1000**G of FIG. **10**G, a bottom surface of the housing structure **114** is then attached to the plurality of housing stopper bumps **120** such that the housing structure **114** and the semiconductor die **104** overlie and are vertically separated from the substrate **102**. After the housing structure **114** is attached, the second plurality of adhesive structures **122** is cured.

With respect to FIG. **10**H, a flowchart **1000**H of some embodiments of a method for forming an IC comprising a stopper bump structure separating a housing structure from a semiconductor die. The IC may, for example, correspond to the IC of FIGS. **10**A-**10**G.

While flowchart **1000**H is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

At **1002**, a plurality of metal housing pads is formed along a lateral inner surface of a housing structure, and a plurality of die stopper bumps are formed on the plurality of metal housing pads. See, for example, FIG. **10**A.

At **1004**, a plurality of housing adhesive structures is formed on and surrounds the plurality of die stopper bumps. See, for example, FIG. **10**B.

At **1006**, a semiconductor die is attached to the plurality of die stopper bumps, and the first plurality of adhesive structures is cured. See, for example, FIG. **10**C.

At **1008**, a frontside surface of the semiconductor die is bonded to the plurality of metal housing pads with a metal wire. See, for example, FIG. **10**D.

At **1010**, a plurality of housing stopper bumps is formed over a substrate. See, for example, FIG. **10**E.

At **1012**, a second plurality of adhesive structures is formed on and surrounds the plurality of housing stopper bumps. See, for example, FIG. **10**F.

At **1014**, a housing structure is attached to the plurality of housing stopper bumps and the second plurality of adhesive structures is cured. See, for example, FIG. **10**G.

With reference to FIGS. **11**A-**11**I, a series of cross sections **1100**A-**1100**I illustrate some embodiments of a method for forming an IC comprising a stopper bump structure separating stacked ICs. The IC may, for example, correspond to the IC of FIG. **3**. Although FIGS. **11**A-**11**I are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **11**A-**11**I are not limited to such a method, but instead may stand alone as structures independent of the method.

As illustrated by the cross-sectional view **1100**A of FIG. **11**A, a substrate **102** is provided. A first plurality of metal pads **110**, a third plurality of metal pads **118**, and a fourth plurality of metal pads **124** are formed over the substrate **102** by the bond pad formation process. A plurality of die stopper bumps **106** is formed over the first plurality of metal pads **110** by the stopper bump formation process. A plurality of housing stopper bumps **120** is formed over the third plurality of metal pads **118** by the stopper bump formation process.

As illustrated by the cross-sectional view **1100**B of FIG. **11**B, a first plurality of adhesive structures **108** is formed, surrounding each of the plurality of die stopper bumps **106**.

As illustrated by the cross-sectional view **1100**C of FIG. **11**C, a semiconductor die **104** is provided. The semiconductor die **104** includes a second plurality of metal pads **116** along a frontside surface of the semiconductor die **104**. A backside surface of the semiconductor die **104** is then attached to the plurality of die stopper bumps **106**. After the semiconductor die **104** is attached, the first plurality of adhesive structures **108** is cured.

As illustrated by the cross-sectional view **1100**D of FIG. **11**D, metal wires **112** are formed, connecting the second plurality of metal pads **116** to the fourth plurality of metal pads **124**, different than the metal pads underlying the plurality of die stopper bumps **106**. In some embodiments, the metal wires **112** are connected to metal pads of the first plurality of metal pads **110** that are separated from the plurality of die stopper bumps **106**.

As illustrated by the cross-sectional view **1100**E of FIG. **11**E, a plurality of metal die pads **310** are formed along a frontside surface of the semiconductor die **104** by the bond pad formation process. In some embodiments, the plurality of metal die pads **310** are formed laterally inside of the second plurality of metal pads **116**. A plurality of overlying die stopper bumps **308** are formed over the plurality of metal die pads **310** by the stopper bump formation process. In some embodiments, the plurality of overlying die stopper bumps **308** are formed with a similar process to that which forms the metal wires **112**.

As illustrated by the cross-sectional view **1100**F of FIG. **11**F, a plurality of overlying adhesive structures **306** is formed over the plurality of metal die pads **310** such that the overlying adhesive structures **306** surround each of the plurality of overlying die stopper bumps **308**. In some embodiments, a single overlying adhesive structure of the plurality of overlying adhesive structures **306** may surround only a single overlying die stopper bump of the plurality of overlying die stopper bumps **308**. In some embodiments, a single overlying adhesive structure of the plurality of overlying adhesive structures **306** may surround more than one die stopper bump of the plurality of overlying die stopper bumps **308**. For instance, the plurality of overlying adhesive structures **306** may be formed by pumping, squeezing, or otherwise providing a liquid around the plurality of overlying die stopper bumps **308** and onto the plurality of metal die pads **310**.

As illustrated by the cross-sectional view **1100**G of FIG. **11**G, a stacked semiconductor die **304** is provided. A backside surface of the stacked semiconductor die **304** is then attached to the plurality of overlying die stopper bumps **308**, such that the stacked semiconductor die **304** overlies the semiconductor die **104**. After the semiconductor die **104** is attached, the plurality of overlying adhesive structures **306** is cured.

As illustrated by the cross-sectional view **1100**H of FIG. **11**H, a second plurality of adhesive structures **122** is formed over the third plurality of metal pads **118** such that the adhesive structures **122** surround each of the plurality of housing stopper bumps **120**.

As illustrated by the cross-sectional view **1100**I of FIG. **11**I, a housing structure is provided. The housing structure **302** is then attached to the plurality of housing stopper bumps **120** such that the housing structure **302** overlies the stacked semiconductor die **304**. After the housing structure **302** is attached, the second plurality of adhesive structures **122** is cured.

With respect to FIG. **11**J, a flowchart **1100**J of some embodiments of a method for forming an IC comprising a stopper bump structure separating stacked ICs. The IC may, for example, correspond to the IC of FIGS. **11**A-**11**I.

While flowchart **1100**J is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

At **1102**, a plurality of die stopper bumps and a plurality of housing stopper bumps are formed on a substrate. See, for example, FIG. **11**A.

At **1104**, a first plurality of adhesive structures is formed on and surrounds the plurality of die stopper bumps. See, for example, FIG. **11**B.

At **1106**, a semiconductor die is attached to the plurality of die stopper bumps, and the first plurality of adhesive structures is cured. See, for example, FIG. **11**C.

At **1108**, a frontside surface of the semiconductor die is bonded to the substrate with a metal wire. See, for example, FIG. **11**D.

At **1110**, a plurality of metal die pads is formed along a frontside surface of the semiconductor die, and a plurality of overlying die stopper bumps is formed over the plurality of metal die pads. See, for example, FIG. **11**E.

At **1112**, a plurality of overlying adhesive structures is formed on and surrounds each of the plurality of overlying die stopper bumps. See, for example, FIG. **11**F.

At **1114**, a stacked semiconductor die is attached to the plurality of overlying die stopper bumps and the plurality of overlying adhesive structures is cured. See, for example, FIG. **11**G.

At **1116**, a second plurality of adhesive structures is formed on and surrounds the plurality of housing stopper bumps. See, for example, FIG. **11**H.

At **1118**, a housing structure is attached to the plurality of housing stopper bumps, and the second plurality of adhesive structures is cured. See, for example, FIG. **11**I.

With reference to FIGS. **12**A-**12**F, a series of cross sections **1200**A-**1200**F illustrate some embodiments of a method for forming an IC comprising a stopper bump structure used to control an angle of a semiconductor die. The IC may, for example, correspond to the IC of FIG. **4**. Although FIGS. **12**A-**12**F are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **12**A-**12**F are not limited to such a method, but instead may stand alone as structures independent of the method.

As illustrated by the cross-sectional view **1200**A of FIG. **12**A, a first plurality of metal pads **110** are formed along a top surface of a provided substrate **102** by the bond pad formation process. In some embodiments, sidewalls of each of the plurality of metal pads **110** contact inner sidewalls of the substrate **102**. A first plurality of die stopper bumps **406** are formed on a first end of the top surface of the substrate **102** by the stopper bump formation process. The first plurality of die stopper bumps **406** directly contacts the first plurality of metal pads **110**. The first plurality of die stopper bumps **406** have a height h**3**. A second plurality of die stopper bumps **402** are formed on a second end of the substrate **102** opposite the first end by the stopper bump formation process. A third plurality of metal pads **118** and a fourth plurality of metal pads **124** are formed over the substrate **102** by the bond pad formation process. A plurality of die stopper bumps **106** is formed over the first plurality of metal pads **110** by the stopper bump formation process. A plurality of housing stopper bumps **120** is formed over the third plurality of metal pads **118** by the stopper bump formation process. The second plurality of die stopper bumps **402** directly contacts the first plurality of metal pads **110**. The second plurality of die stopper bumps **402** have a height h**4**. In some embodiments, h**4** is larger than h**3**. In some embodiments, the plurality of housing stopper bumps **120** may each have heights such that their top surfaces define a plane that is tilted relative to the top surface of the substrate **102**.

As illustrated by the cross-sectional view **1200**B of FIG. **12**B, a first plurality of die adhesive structures **408** is formed, surrounding each of the first plurality of die stopper bumps **406** such that the first plurality of die adhesive structures **408** has an approximate height equal to h**3**. A second plurality of die adhesive structures **404** is formed, surrounding each of the second plurality of die stopper bumps **402** such that the second plurality of die adhesive structures **404** has an approximate height equal to h**4**. For instance, the first plurality of die adhesive structures **408** and the second plurality of die adhesive structures **404** may be formed by pumping, squeezing, or otherwise providing a liquid around the first plurality of die stopper bumps **406** and the second plurality of die stopper bumps **402**, respectively, and onto the plurality of metal pads **110**.

As illustrated by the cross-sectional view **1200**C of FIG. **12**C, a semiconductor die **104** is provided. The semiconductor die **104** includes a second plurality of metal pads **116** along a frontside surface of the semiconductor die **104**. A backside surface of the semiconductor die **104** is then attached to the first plurality of die stopper bumps **406** and the second plurality of die stopper bumps **402**, such that the semiconductor die **104** has a tilt angle **A1** with respect to a top surface of the substrate **102**. After the semiconductor die **104** is attached, the first plurality of die adhesive structures **408** and the second plurality of die adhesive structures **404** are cured.

As illustrated by the cross-sectional view **1200**D of FIG. **12**D, metal wires **112** are formed, connecting the second plurality of metal pads **116** to the fourth plurality of metal pads **124**, different than the metal pads underlying the first plurality of die stopper bumps **406** and the second plurality of die stopper bumps **402**. In some embodiments, the metal wires **112** are connected to metal pads of the first plurality of metal pads **110** that are different than the metal pads underlying the plurality of die stopper bumps **106**.

As illustrated by the cross-sectional view **1200**E of FIG. **12**E, a second plurality of adhesive structures **122** is formed over the third plurality of metal pads **118** such that the adhesive structures **122** surround each of the plurality of housing stopper bumps **120**.

As illustrated by the cross-sectional view **1200**F of FIG. **12**F, a housing structure is provided. The housing structure **114** is then attached to the plurality of housing stopper bumps **120** such that the housing structure **114** overlies the semiconductor die **104**. After the housing structure **114** is attached, the second plurality of adhesive structures **122** is cured.

With respect to FIG. **12**G, a flowchart **1200**G of some embodiments of a method for forming an IC comprising a stopper bump structure used to control an angle of a semiconductor die. The IC may, for example, correspond to the IC of FIGS. **12**A-**12**F.

While flowchart **1200**G is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

At **1202**, a first plurality of die stopper bumps is formed on a substrate at a first height, a second plurality of die stopper bumps is formed on the substrate at a second height, and a plurality of housing stopper bumps is formed on the substrate. See, for example, FIG. **12**A.

At **1204**, a first plurality of die adhesive structures is formed on and surrounds the first plurality of die stopper bumps at the first height, and a second plurality of die adhesive structures is formed on and surrounds the second plurality of die stopper bumps at the second height. See, for example, FIG. **12**B.

At **1206**, a semiconductor die is attached to the first and second pluralities of die stopper bumps, and the first and second pluralities of adhesive structures are cured. See, for example, FIG. **12**C.

At **1208**, a frontside surface of the semiconductor die is bonded to the substrate with a metal wire. See, for example, FIG. **12**D.

At **1210**, a second plurality of adhesive structures is formed on and surrounds the plurality of housing stopper bumps. See, for example, FIG. **12**E.

At **1212**, a housing structure is attached to the plurality of housing stopper bumps, and the second plurality of adhesive structures is cured. See, for example, FIG. **12**F.

With reference to FIGS. **13**A-**13**E, a series of cross sections **1300**A-**1300**E illustrate some embodiments of a method for forming an IC comprising an electrically conductive stopper bump structure. The IC may, for example, correspond to the IC of FIG. **5**. Although FIGS. **13**A-**13**E are described in relation to a method, it will be appreciated that the structures disclosed in FIGS. **13**A-**13**E are not limited to such a method, but instead may stand alone as structures independent of the method.

As illustrated by the cross-sectional view **1300**A of FIG. **13**A, a substrate **102** is provided. A first plurality of metal pads **110** and a third plurality of metal pads **118** are formed on a top surface of the substrate **102**. A plurality of die stopper bumps **106** is formed on a top surface of the first plurality of metal pads **110** by the stopper bump formation process. A plurality of housing stopper bumps **120** is formed on a top surface of the third plurality of metal pads **118** by the stopper bump formation process.

As illustrated by the cross-sectional view **1300**B of FIG. **13**B, a first plurality of adhesive structures **108** is formed, surrounding each of the plurality of die stopper bumps **106**.

As illustrated by the cross-sectional view **1300**C of FIG. **13**C, a semiconductor die **104** is provided. A plurality of metal die pads **502** are formed along a frontside surface of the semiconductor die **104** by the bond pad formation process, such that a first surface of the plurality of metal die pads **502** contacts the semiconductor die **104**. A second surface of the plurality of metal die pads **502** opposite that of the first surface is then attached to the plurality of die stopper bumps **106** to establish an electrical connection between the first plurality of metal pads **110** and the plurality of metal die pads **502**. After the semiconductor die **104** is attached, the first plurality of adhesive structures **108** is cured.

As illustrated by the cross-sectional view **1300**D of FIG. **13**D, a second plurality of adhesive structures **122** is formed over the third plurality of metal pads **118** such that the adhesive structures **122** surround each of the plurality of housing stopper bumps **120**.

As illustrated by the cross-sectional view **1300**E of FIG. **13**E, a housing structure is provided. The housing structure **114** is then attached to the plurality of housing stopper bumps **120** such that the housing structure **114** overlies the stacked semiconductor die **304**. After the housing structure **114** is attached, the second plurality of adhesive structures **122** is cured.

With respect to FIG. **13**F, a flowchart **1300**F of some embodiments of a method for forming an IC comprising an electrically conductive stopper bump structure. The IC may, for example, correspond to the IC of FIGS. **13**A-**13**E.

While flowchart **1300**F is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

At **1302**, a plurality of die stopper bumps and a plurality of housing stopper bumps are formed on a substrate. See, for example, FIG. **13**A.

At **1304**, a first plurality of adhesive structures is formed on and surrounds the plurality of die stopper bumps. See, for example, FIG. **13**B.

At **1306**, a plurality of metal die pads is formed along a frontside surface of a semiconductor die, the plurality of metal die pads is attached to the plurality of die stopper bumps, and the first plurality of adhesive structures is cured. See, for example, FIG. **13**C.

At **1308**, a second plurality of adhesive structures is formed on and surrounds the plurality of housing stopper bumps. See, for example, FIG. **13**D.

At **1310**, a housing structure is attached to the plurality of housing stopper bumps, and the second plurality of adhesive structures is cured. See, for example, FIG. **13**E.

FIG. **14** illustrates a cross-sectional view **1400** of some embodiments of an IC comprising a stopper bump structure. The IC may be, for example, the IC of FIG. **1**. The IC comprises a first plurality of metal pads **110** overlying a substrate **102**. A first plurality of stopper bumps **1404***a* overlies one of the first plurality of metal pads **110** and separates the substrate **102** from an overlying semiconductor die **104**. A first plurality of stopper bumps **1404***a* overlies one of the first plurality of metal pads **110** and further separates the substrate **102** from the overlying semiconductor die **104**. In some embodiments, each stopper bump of the first and second pluralities of stopper bumps **1404***a* and **1404***b* may have a different height, as a formation process of the first and second pluralities of stopper bumps **1404***a* and **1404***b* may have a standard deviation in height. In some embodiments, the standard deviation may range from 3 um to 5 um. In some embodiments, a stopper bump of the first plurality of stopper bumps **1404***a* with a greatest height directly contacts the semiconductor die **104** and a stopper bump of the second plurality of stopper bumps **1404***b* with a greatest height directly contacts the semiconductor die **104**, such that the stopper bump with the greatest height of the first plurality of stopper bumps **1404***a* and the stopper bump with the greatest height of the second plurality of stopper bumps **1404***b* determine a distance of the semiconductor die **104** from the substrate **102**.

Accordingly, in some embodiments, the present disclosure relates to an integrated chip (IC), comprising a substrate, a first die disposed over the substrate, a metal wire attached to a frontside of the first die, and a first plurality of die stopper bumps disposed along a backside of the first die and configured to control an angle of operation of the first die. The first plurality of die stopper bumps directly contacts a backside surface of the first die.

In other embodiments, the present disclosure relates to a method for forming an integrated chip (IC), comprising forming a first stopper structure on a substrate, forming a first adhesive structure on and surrounding the first stopper structure, attaching a first die to the first adhesive structure and curing the first adhesive structure, and bonding a bottom surface of the first die to the substrate with a metal wire.

In yet other embodiments, the present disclosure relates to an integrated chip (IC), comprising a substrate including a plurality of metal pads on a top surface of the substrate, a die disposed over the substrate, and the die including a first side over the top surface of the substrate and a second side over the first side, a housing structure disposed over the second side of the die and surrounding sidewalls of the die, a plurality of stopper bumps disposed between the first side of the die and a first metal pad of the plurality of metal pads, and a plurality of adhesive structures disposed on and surrounding each of the plurality of stopper bumps.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An integrated chip (IC), comprising:
a substrate;
a die disposed vertically over the substrate;
a housing structure disposed over the substrate and the die;
a metal pad disposed over the substrate and coupled to an underside of the housing structure, wherein the metal pad has an outer sidewall that extends horizontally past an outer sidewall of the die;
an adhesive coupling the die to the metal pad; and
stopper bumps laterally surrounded by the adhesive, the stopper bumps contacting the die and the metal pad.

2. The IC of claim **1**, further comprising:
a metal wire extending between the substrate and a frontside of the first die.

3. The IC of claim **1**,
wherein the housing structure surrounds sidewalls of the die.

4. The IC of claim **3**, further comprising:
housing stopper bumps directly contacting and disposed between the housing structure and the substrate.

5. The IC of claim **4**, wherein the housing stopper bumps and the stopper bumps comprise the same material as one another.

6. The IC of claim **3**, further comprising:
a cavity arranged directly between the housing structure and an uppermost surface of the die and separating the housing structure from the uppermost surface of the die.

7. The IC of claim **1**, further comprising:
a metal wire extending from the metal pad to a first side of the die, wherein the first side of the die faces the substrate.

8. An integrated chip (IC), comprising:
a substrate;
a die disposed over the substrate;
a housing structure disposed over the die and surrounding sidewalls of the die; and
a plurality of die stopper bumps arranged as a plurality of groups of die stopper bumps, wherein each group of the plurality of groups of die stopper bumps is disposed between an upper surface of the die and a lower surface of the housing structure, wherein the plurality of die stopper bumps suspend the die from the housing structure over the substrate.

9. The IC of claim **8**, further comprising:
a plurality of metal pads disposed along a lateral portion of an inner surface of the housing structure, and
a metal wire extending from a metal pad of the plurality of metal pads to the die.

10. The IC of claim 8, further comprising:
a first plurality of adhesive structures, wherein each of the first plurality of adhesive structures is disposed on and surrounds a corresponding group of the plurality of groups of die stopper bumps.

11. The IC of claim 8, wherein the plurality of die stopper bumps are surrounded by an adhesive.

12. The IC of claim 11, further comprising:
a plurality of metal pads disposed along a lateral portion of an inner surface of the housing structure, and
a metal wire extending from a metal pad of the plurality of metal pads to the die.

13. The IC of claim 12, wherein each die stopper bump of a group of die stopper bumps and the metal wire contact the metal pad.

14. An integrated chip (IC), comprising:
a substrate having an upper surface and a lower surface;
a dome-like housing structure disposed over the substrate, wherein an interior surface of the dome-like housing structure and the upper surface of the substrate define a cavity; and
a die coupled to the interior surface of the dome-like housing structure and thereby suspended over an upper surface of the substrate, the die having a lower surface proximate to the substrate and an upper surface over the lower surface of the die;
an adhesion structure sandwiched between the upper surface of the die and the interior surface of the dome-like housing structure, the adhesion structure having a volume defined by an outer sidewall of the adhesion structure; and
multiple die stopper bumps each enveloped by the adhesion structure, the multiple die stopper bumps each having a lower surface contacting the upper surface of the die and each having an upper surface contacting the interior surface of the dome-like housing structure.

15. The IC of claim 14, further comprising:
a plurality of housing stopper bumps directly contacting and disposed between a bottom surface of the dome-like housing structure and the upper surface of the substrate.

16. The IC of claim 14, wherein each of the multiple die stopper bumps comprises a solder ball bump.

17. The IC of claim 14, wherein each of the multiple die stopper bumps comprises a stud bump, wherein the stud bump comprises a lower portion and an upper portion, wherein the lower portion has a maximum width greater than that of the upper portion, and wherein the lower portion comprises sidewalls laterally offset from sidewalls of the upper portion.

18. The IC of claim 14, wherein the multiple die stopper bumps are separated into at least one group of die stopper bumps, wherein each group of die stopper bumps comprises at least one row of die stopper bumps and at least one column of die stopper bumps.

19. The IC of claim 18, wherein the adhesive structure is a single body that surrounds an outer perimeter of the group of die stopper bumps and that resides between die stopper bumps of the group of die stopper bumps to adhere the die stopper bumps of the group of die stopper bumps to one another.

20. The IC of claim 14, further comprising:
a metal wire extending between a first contact or bond pad on the lower surface of the die and a second contact or bond pad on the interior surface of the dome-like housing structure.

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